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Design and Prototyping of a User-centered Consent Flow

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Abstract

Within the current digital environment, large digital platforms collect and process extensive amounts of user data, raising ethical concerns about privacy, transparency, and meaningful user control. Existing consent mechanisms are perceived as confusing and opaque, often failing to provide genuine control. This thesis examines how a user-centered consent flow can be designed and prototyped in a way that promotes transparency, meaningful user control, and scalable, privacy-conscious data sharing. Through qualitative analysis of existing solutions and ethical principles, a prototype consent flow aligning with dynamic consent principles and regulatory guidelines is designed and implemented. The prototype features structured interfaces (purpose categorization, dropdown controls), a centralized dashboard for managing consent, and a history view for transparency and accountability. Evaluation combines criteria-based assessment with user testing involving fifteen participants.

User testing demonstrates strong performance across the evaluated criteria, with participants particularly valuing the dashboard used for reviewing and modifying past decisions and nearly universally agreeing that manipulative design was avoided. However, scalability emerges as a significant challenge, rooted fundamentally in cognitive load rather than interface organization, as participants perceive the prototype as increasingly difficult to manage and understand as the number of data categories, processing purposes, and third-party actors increases, even with hierarchical structures. The prototype operates in a simplified environment with mock data, yet validates feasibility. Evaluation is limited to criteria-based assessment and convenience sampling. Despite these constraints, the work shows that user-centered consent flows can promote transparency and meaningful user control through lifecycle-based consent design, while also revealing that addressing scalability requires moving beyond interface design to systemic data practices.

Keywords

Consent flow, User-centered consent, Dynamic consent, Data privacy, Privacy, User control, Ethical Data, Transparency

Sammanfattning

I dagens digitala miljö samlar och bearbetar stora digitala plattformar omfattande mängder användardata, vilket väcker etiska frågor om integritet, transparens och meningsfull användarkontroll. Befintliga samtyckesmekanismer upplevs som förvirrande och otydliga och ger ofta inte verklig kontroll. Detta examensarbete undersöker hur det går att utforma och skapa en prototyp för ett användarcentrerat samtyckesflöde för att främja transparens, meningsfull användarkontroll och skalbar, integritetsmedveten datadelning. Genom kvalitativ analys av befintliga lösningar och etiska principer designas och implementeras ett prototypiskt samtyckesflöde i linje med principerna för dynamiskt samtycke och regulatoriska riktlinjer. Prototypen innehåller strukturerade gränssnitt (ändamålskategorisering, dropdown-kontroller), en centraliserad instrumentpanel för hantering av samtycke och en historikvy för transparens och ansvarighet. Utvärderingen kombinerar kriteriebaserad bedömning och användartester med femton deltagare.

Användartesterna visar goda resultat i de utvärderade kriterierna. Deltagarna uppskattade särskilt instrumentpanelen för att granska och ändra tidigare beslut och var nästan eniga om att manipulativ design hade undvikits. Skalbarhet framträder dock som en betydande utmaning, grundad främst i kognitiv belastning snarare än i gränssnittsorganisation. Deltagarna upplevde prototypen som svårare att hantera och förstå ju fler datakategorier, ändamål för behandling och tredjeparter som tillkom, även vid användning av hierarkiska strukturer. Prototypen körs i en förenklad miljö med simulerade data, men validerar genomförbarheten. Utvärderingen är begränsad till kriteriebaserad bedömning och bekvämlighetsurval. Trots dessa begränsningar visar arbetet att användarcentrerade samtyckesflöden kan främja transparens och meningsfull användarkontroll genom livscykelbaserad samtyckesdesign, samtidigt som resultaten visar att hanteringen av skalbarhetsutmaningar kräver mer än enbart gränssnittsdesign och även måste omfatta systemiska datapraxis.

Nyckelord

Samtyckesflöde, Användarcentrerat samtycke, Dynamiskt samtycke,
Dataskydd, Integritet, Användarkontroll, Etisk datahantering, Transparens

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AI	Artificial Intelligence
GDPR	General Data Protection Regulation
PbD	Privacy by Design
SDG	Sustainable Development Goal
SPA	Single Page Application

Chapter 1

Introduction

In modern society, people heavily rely on the Internet and the use of its service platforms and social media [1][2]. These platforms, owned and operated by large corporations, collect and analyze extensive amounts of user data. By monitoring users' digital footprints, this data is leveraged to predict and influence preferences and behavior. As a result, user data has become highly valuable for large data driven companies [3]. The rapid development of Artificial Intelligence (AI) has further enabled the use of data to predict what users are likely to engage with or want to see online [4].

This development has spurred research within the field of ethical data, which concerns the moral principles of the collection, storage, and use of people's personal data while upholding values of fairness, privacy, and accountability [5].

1.1 Background

Digital service platforms functions as online hubs where diverse actors interact and exchange value [6]. However, these platforms increasingly rely on centralized and surveillance-driven data practices to facilitate personalized services [3]. These practices often involve the collection, processing, and sharing of data with limited transparency for users. As a result, users are often unaware and have little control over how their data is used [7]. In this context, meaningful consent mechanisms are crucial, as they enable users to approve, deny, or modify how platforms collect and process their personal data [8].

On digital service platforms, it is common for user data to be exchanged among multiple parties. Users often provide personal information, usage data, or behavioral signals to a service provider to access services. However,

this personal data may subsequently be shared with third parties beyond the customer's original expectations, raising concerns regarding privacy, transparency, and control [9].

Consent flows are structured interaction mechanisms that can be understood as tools for consent management, designed to organize and manage the lifecycle of user consent across multiple interactions. Building on this definition, consent flows specify what data is shared, with which parties, for what purposes, and under what conditions. Consent flows are meant to enable users to grant, modify, or withdraw consent over time, thereby supporting privacy through informed decision-making and transparent data governance. These mechanisms should support user control, including the ability to revoke, and to adjust preferences as needs change [10]. To further strengthen user protections, they should support time limited consent and enhance transparency by clearly communicating how personal data is collected and processed, in accordance with dynamic consent principles [11]. Meaningful consent mechanisms are therefore essential for addressing the power imbalances inherent in digital platforms, where users have limited visibility into and control over their personal data.

Additionally, consent mechanisms must be scalable, ensuring usability and enforceability across platforms with large user bases and complex data exchanges [12]. Recent research has explored decentralized and privacy-preserving architectures, such as blockchain-based platforms, as potential means to give users more control over their data. While promising, these approaches introduce practical challenges that affect usability and scalability [13]. Together, these aspects form a conceptual framework for understanding meaningful consent systems and the key design considerations they entail.

1.2 Problem

Despite their intended purpose, existing consent mechanisms are often complex, opaque, and poorly aligned with users' needs and understanding. This is often achieved through dark patterns, which are design strategies that manipulate or nudge users into making unintended choices [7]. This results in consent that is largely nominal and primarily serves legal compliance due to regulatory loopholes [14]. Consequently, many current consent frameworks fail to provide meaningful user control, undermining privacy protection and ethical data use.

Although decentralized, privacy-preserving approaches (e.g., blockchain) can enhance transparency and user control, they face practical challenges such as limited scalability, integration difficulties, and high computational overhead [13][15]. This reduces their applicability across various service platforms. As a result, there is still limited understanding of how user-centered consent flows can be crafted to support privacy-preserving and scalable data sharing while ensuring meaningful user control. This gap has motivated this thesis, which seeks to answer the following research question: How can a user-centered consent flow be designed and implemented to promote transparency and meaningful user control regarding data sharing within a digital service platform?

1.3 Purpose

The purpose of this thesis is to advance meaningful user control and transparency in digital data sharing. Through this work, this thesis seeks to contribute to more ethical and user-centered approaches to data governance in increasingly complex digital ecosystems.

1.4 Goal

The goal of this thesis is to investigate, design, and prototype a user-centered consent flow that supports transparency and meaningful user control in digital data sharing at scale.

1.5 Risks and Benefits

A key risk in developing consent mechanisms of this nature is the potential for increasing complexity, which can result in information overload. Overly complex consent flows may lead to consent fatigue, a phenomenon in which users give consent without fully reading or understanding the information presented [16]. The challenge, therefore, lies in maximizing transparency while ensuring simplicity in the design.

The potential benefits of the project are significant. The developed consent flows can provide users with greater autonomy over data exchange, enabling them to exercise full control over their personal information. Furthermore, successfully implementing meaningful user-centered consent mechanisms

within a digital service platform can enhance user trust and in turn, increase engagement and willingness to interact with these platforms.

At a societal level, promoting meaningful consent through research and the development of effective consent mechanisms could contribute to broader norms that favor privacy-conscious digital interactions.

1.6 Ethics and Sustainability

This thesis is situated within the field of ethical data, and involves several critical ethical considerations. A mapping study from 2021 has shown that a majority of primary studies in mobile app privacy are focused on methods for leak detection with the least amount of studies focused on studying user feedback regarding privacy concerns [17]. While leaks are important, a well designed consent flow can help users regain control over what data they are willing to share.

When developing consent flows, it is essential to avoid manipulative design practices; instead, the design should support informed and voluntary decision-making. Among the existing consent mechanisms used today, manipulative designs and gamification are strategies that are commonly used to steer users toward accepting consent [18][19][20][21]. Research demonstrates that the impact of these designs is not uniformly distributed. Studies of users in vulnerable positions show that vulnerable populations, including those with lower digital literacy, non-native speakers and teenagers experience disproportionate difficulty in recognizing and resisting manipulative design patterns [22][23]. Research specifically examining teenagers interactions with manipulative designs reveals how their personal and social contexts shape their vulnerability to these practices [23].

Vulnerability is systematically exploited through economic mechanisms. Platforms actively profile economically vulnerable users and deliberately serve them manipulative designs exploiting their financial situation [22] while cookie paywalls tie privacy rights to ability to pay [24]. Even users who explicitly attempt to minimize their spending or to protect their privacy remain susceptible to strategically placed economic incentives and choice architecture manipulation, resulting in measurable economic and privacy harm [22]. This creates systematic inequality in the ability to protect privacy: wealthier users can more easily afford to opt out or ignore incentives, while lower-income users face designs specifically engineered to exploit their economic constraints [22].

Beyond economic barriers, accessibility barriers further limit meaningful consent. Privacy policies have since 2001 greatly increased in length and complexity of language, becoming more ambiguous [25]. This creates barriers not just for average users, but particularly for vulnerable populations: non-native speakers, users with lower literacy levels, elderly users, and those with cognitive disabilities or limited time. Research shows that even when vulnerable users attempt to understand consent information, they may not be able to do so independently, requiring help from others and ultimately making consent decisions without comprehension [22].

Critically, this creates a self-reinforcing cycle: when users experience harm from manipulative designs, they may disengage from digital systems altogether, deepening their digital exclusion and reinforcing existing inequalities [22]. This positions design itself as a lever for either perpetuating or mitigating inequalities. Well-designed consent mechanisms that prioritize accessibility, clarity, and meaningful user control can help users remain engaged with digital services while protecting their privacy and autonomy, whereas poorly designed mechanisms actively worsen inequality by excluding those already in vulnerable positions.

These concerns relate directly to the United Nations Sustainable Development Goals (SDGs). SDG 10 ”Reduce inequality within and among countries” [26] is tied to digital inequality created by manipulative design practices and inaccessible consent mechanisms. This exemplifies how technological systems can perpetuate or worsen socioeconomic disadvantage. Additionally, SDG 16 ”Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels” [26] emphasizes the importance of transparency, accountability, and the protection of fundamental rights. Meaningful consent flows are mechanisms through which institutions can be held accountable to users, and through which transparency and user agency can be strengthened in the digital ecosystem.

Consent flows act as mechanisms that facilitate the collection and management of personal data [27]. However, they face significant limitations. Overall trust in existing consent flows is low, as over 90% of users believe that some applications share their data without consent [28]. This view is supported by evidence that many applications indeed share data in ways not consistent with their privacy policies [29][30]. A well implemented and secure consent flow may be able to guarantee that data is only shared with a third party with consent, but it cannot guarantee that the third party won't use it for other purposes than stated or share it with additional actors. For an effective consent flow it is essential that the information is clear and brief enough to remain effective. Consequently, the feasibility of implementing such user-centered consent flows in an industrial setting depends on the adoption of measures that ensure secure data transmission, processing, and storage. These measures should be supported and reinforced by regulatory frameworks that safeguard data integrity and hold data-driven companies accountable.

The results of this thesis, particularly the development of a user-centered consent flow, may be of interest to digital platforms that collect and process large amounts of user data. The environmental implications of maintaining extensive data infrastructures must therefore be acknowledged. This relates to SDG 12 and 13, which emphasize sustainable resource management and carbon emissions reduction [26]. In industrial contexts, storing and processing significant volumes of data can lead to considerable energy consumption and carbon dioxide emissions [31]. A user-centered approach to consent that enables users to control what data is collected may also contribute indirectly to more responsible data consumption by reducing the collection of unnecessary or unwanted data. Consequently, the development and implementation of consent flows should not compromise existing systems in terms of energy efficiency.

1.7 Methods

This project is conducted using a qualitative research method. Qualitative research focuses on understanding meanings, opinions, and behaviors in order to support the development of a digital system [32]. This differs from quantitative research methods, which emphasize experiments and testing by measuring variables to verify or falsify a hypothesis. In addition, this project follows an applied research approach, in which existing research is utilized to address a practical problem [32].

After developing the user-centered consent flow prototypes, a criteria-based evaluation of the system is conducted. In this approach, the evaluation is guided by predefined standards, or criteria, such as checklists, heuristics, principles, and quality ideals [33]. The criteria are theory-driven, meaning they are grounded in established frameworks [33]. In this project, the criteria are derived from principles of dynamic consent, Privacy by Design (PbD), and relevant regulatory policies [34]. The purpose of the evaluation is to assess the prototypes from a user-centered perspective. Focusing on these specific criteria allows the evaluation to concentrate on the qualities that matter most for user-centered consent flows, while deliberately placing less emphasis on aspects that are less critical in this context.

1.8 Stakeholders

The project is conducted in collaboration with Fixmeapp AB, which is developing a social booking platform emphasizing privacy and trust [35]. One of their key research areas focuses on how the platform can provide meaningful, transparent, and user-driven control over personal data. This includes exploring how consent can be structured as granular and revocable tokens, as well as how a user-centered consent interface could be designed [36]. By investigating, designing, and prototyping a user-centered consent flow, the outcomes of this research will contribute to Fixmeapp's knowledge and support their ongoing efforts to implement transparent and meaningful consent mechanisms.

1.9 Delimitations

This project focuses solely on the design and development of a user-centered consent flow within a centralized architectural environment that simulates a digital service platform. A centralized architecture is chosen because it reflects the standard model used by most commercial service platforms and simplifies implementation [37][38]. While decentralized approaches like blockchain can enhance privacy and control, they introduce significant scalability, integration, and computational challenges [13][15] that are beyond the scope of this thesis. The work concentrates on implementing consent management mechanisms throughout the consent lifecycle, without aiming to develop a fully functioning service platform. Specifically, the work addresses interaction patterns, user decisions, information presentation, and the enforcement of user consent and control over data sharing. All data exchanged within the simulated environment is mock data and has no relation to the project stakeholders.

The user-centered consent flow is designed to be privacy-conscious by adhering to Privacy by Design (PbD) principles [34]. However, the design and prototype exclude data security mechanisms, such as cryptographic protocols and their implementation. The project operates under the assumption that data integrity and confidentiality are inherent properties of the platform. Consequently, privacy-preserving data exchange pipelines and privacy-oriented database design are not examined. While scalability regarding information presentation to the user is considered, large-scale back-end data processing falls outside the scope of this work. The prototype serves as a proof-of-concept for user-centered consent flows.

The prototype is evaluated through a criteria-based evaluation process. In addition, user testing is conducted on a small scale using convenience sampling due to time constraints.

1.10 Outline

Chapter 2 presents relevant background information on consent flows. Chapter 3 outlines the methods used throughout the work. Chapter 4 presents key insights from the literature review and the developed design criteria. Chapter 5 describes the overall design of the consent flow and the implemented prototype. Chapter 6 presents the criteria-based evaluation and user testing of the prototype. Chapter 7 concludes the thesis with a discussion and directions for future work.

Chapter 2

Background

This chapter presents the theoretical foundation of the project. Section 2.1 provides an overview of digital service platforms and the data ecosystems in which they operate. Section 2.2 introduces existing consent mechanisms and principles, including their strengths and limitations. Section 2.3 discusses identified consent interface design patterns. Section 2.4 introduces the concept of dynamic consent. Section 2.5 outlines the regulatory frameworks that govern consent flows. Section 2.6 presents key privacy principles relevant to the design of software systems. Section 2.7 examines the scalability challenges associated with consent management in complex digital environments. Section 2.8 discusses relevant development approaches and technologies for implementing a web application prototype. Finally, Section 2.9 reviews existing research related to user-centered consent flows.

2.1 Digital Service Platforms and Data Ecosystems

Digital service platforms serve as the dominant infrastructure for many online services. These platforms operate as meeting places for multiple actors, including consumers, advertisers, developers, and third-party service providers [6]. In this context, they form part of broader data ecosystems, understood as interconnected networks of actors, technologies, and processes involved in the collection, exchange, and use of data across organizational boundaries [39]. Platforms generate value by facilitating interactions among

these groups while simultaneously collecting and leveraging relevant user data [39]. Through this process, platforms benefit from network effects: as more users participate and share data, the overall utility and attractiveness of the platform increases [40]. Recent research further extends this perspective by introducing the concept of Artificial Intelligence (AI) driven data network effects, where the use of AI amplifies the value derived from accumulated data. In such settings, continuous learning from user data enhances personalization, service quality, thereby strengthening competitive advantages for platform providers [40].

Due to this, user data plays an essential role within digital service platforms in enabling personalization, optimization, and monetization. In light of this, modern platform business models can be characterized as inherently data-driven. Recent analyses suggest that large technology companies have moved beyond collecting data solely to improve their services, instead leveraging it for predictive analytics, targeted advertising, behavioral influence, and data commercialization [3].

The multi-dimensional use of data contributes to the emergence of complex data ecosystems in which multiple actors participate in extended data processing chains. When several actors are involved in data processing activities, determining who processes data, for what purposes, and under which conditions becomes increasingly difficult [41]. Consequently, consent management within digital service platforms is not merely a legal or interface-level challenge, but also an architectural and ecosystem-level concern.

2.2 Consent Mechanisms and Consent Flows

Within current data protection frameworks, consent is one of the most foundational legal bases for processing personal data [42][43]. Under the European Union's General Data Protection Regulation (GDPR), consent is defined as a freely given, specific, informed, and unambiguous indication that the data subject has agreed to the processing of their personal data [42]. This definition underscores that consent is not merely a formality, but a structured mechanism designed to safeguard individual autonomy within digital environments.

Dissecting this definition reveals several key requirements. Consent must be informed, meaning the user understands what data is being processed, for what purpose, and by which parties [42]. It must be specific, indicating that all processing activities are clearly explained rather than left vague or open to unspecified future uses [42]. Consent must also be freely given, without coercion or manipulation [42]. Finally, consent must be revocable, ensuring that the process of withdrawal is as straightforward and accessible as giving consent [42].

These requirements provide a normative foundation for how consent should be processed, which keeps data collection transparent. However, the practical implementation of such consent mechanisms presents significant challenges [7][10]. As discussed in Section 2.1, data processing occurs within complex ecosystems involving multiple actors rather than through isolated two-party exchanges. In these environments, studies have shown that consent may be provided via a single interface but subsequently shared with multiple third parties, often with limited transparency [44][45]. As a result, the informational requirements for these mechanisms become increasingly complex.

The concept of informed consent extends beyond legal compliance to encompass usability and cognitive considerations. Many current consent mechanisms present users with extensive privacy policies, layered notices, and complex technical descriptions that exceed realistic expectations of comprehension [14]. This often results in formal transparency through disclosure, but limited actual understanding by the user. Furthermore, research has shown that coercive, manipulative, and nudging strategies have been used in interface designs, leading to compliance that may not reflect the user's true preferences [14][7]. These strategies are commonly referred to as dark patterns, and include preselected options, highlighted buttons, and the concealment of privacy settings [14]. In such cases, consent mechanisms fail to provide users with meaningful control.

As a result, usability is a critical consideration when developing consent mechanisms. Research has identified seven key usability requirements for effective consent interfaces: user needs, user ability and effort, user awareness, user comprehension, user sentiment, decision reversal, and nudging patterns [46][47]. These requirements provide a structured framework to ensure that consent interfaces are not only legally compliant but also understandable, manageable, and respectful of users' autonomy. Understanding how these principles manifest in actual interface design requires examining empirical research on consent interface patterns.

2.3 Consent Interface Design Patterns

User preferences for consent interface design have been systematically evaluated through comparative studies examining real-world cookie consent notices deployed across major websites. An in-the-wild study with 98 participants ranking five popular consent notice designs from major Consent Management Platforms identified distinct patterns in user preference and usability outcomes [48]. The five designs examined were: (1) ”Slider Option”, that allows users to select from three predefined cookie consent levels (required only, required and functional, or required along with functional and advertising); (2) ”Cookie Categories”, allowing users to individually select or deselect each cookie category, with required cookies remaining non-negotiable; (3) ”Binary Choice”, providing a single ”Continue” button emphasizing default acceptance; (4) ”Full Vendor List”, which enables granular control over individual cookies and tracking entities with a ”Reject All” option; and (5) ”Full Vendor List without Reject All”, identical to the previous design but without the rejection shortcut [48].

Across these designs, the slider design was identified as statistically significantly preferred over alternatives [48]. Beyond design preferences, this work identified five key usability factors driving user satisfaction: ease of use, amount of information, customizability, decision-making time, and clarity/transparency [48]. The study also revealed critical insights about user decision-making in consent contexts: while users generally select privacy-preserving options when presented with clear choices, many do not fully understand the implications of their decisions, with 71% of participants incorrectly believing they would still be tracked even after declining consent [48]. Additionally, the research demonstrates the substantial influence of default options on user behavior, with 30-40% of users accepting default settings across multiple design variants [48].

Interactive design approaches further demonstrate that the interaction modality itself significantly influences user engagement and awareness of data disclosures. A comparative study of checkbox selections, drag-and-drop interactions, and swiping gestures shows that while traditional checkboxes enable faster interactions, alternative design patterns such as drag-and-drop and swiping mechanisms generate higher levels of user attentiveness to data flows and greater user satisfaction with the consent experience [49]. These findings suggest that interaction design plays a critical role in helping users achieve genuine informed consent rather than merely completing a consent task, aligning with the broader principle that consent interfaces should be designed not only for efficiency but also for meaningful user engagement with the information being disclosed.

2.4 Dynamic Consent

To provide meaningful user control, consent management is frequently mentioned as a key requirement for addressing these issues [50]. The term dynamic consent refers to an approach to informed consent that addresses the limitations of static consent mechanisms [11]. Static consent is typically a one-time, non-changeable agreement, which limits users' ability to manage their data over time [11]. In contrast, dynamic consent considers consent as part of an ongoing lifecycle rather than a single event [11][51]. This lifecycle encompasses not only the initial granting of consent but also the potential for modification or withdrawal. By adopting this approach, dynamic consent enables users to maintain comprehensive control over the flow of their personal data [51].

The development of dynamic consent has enabled greater granularity in consent design, which is considered a central aspect of effective consent management [52][50]. Granular consent enhances user control by allowing individuals to provide consent for specific data fields or processing purposes rather than broad, undifferentiated categories [53]. However, increased granularity also introduces challenges, as more detailed consent options may add interface complexity and contribute to cognitive overload [53]. If implemented excessively, such fragmentation can lead to consent fatigue, a phenomenon in which repeated or overly detailed consent requests reduce users' willingness or ability to meaningfully evaluate their choices [16][54].

2.5 Regulatory Framework for Data Protection

Consent mechanisms are shaped not only by usability but also by legal requirements. Regulatory frameworks such as the European Union's General Data Protection Regulation (GDPR) define the conditions under which consent is considered valid, including requirements that it be freely given, specific, informed, and capable of being withdrawn, thereby imposing restrictions on how user permissions are collected, processed, and stored [55]. This extends beyond interface-level design, providing overarching principles that guide how consent mechanisms should be implemented in digital systems.

Under GDPR, there exists foundational principles of how lawful data processing should be conducted [55]. Central among these includes lawfulness, fairness, and transparency, purpose limitation, data minimization, accuracy, storage limitation, integrity and confidentiality, and accountability [56]. Together these form the principles organizations must adhere to when communicating the use of the collected data and enforcing dynamic consent management [11].

As discussed in Section 2.2, all consent must be freely given, specific, informed, and unambiguous. This requires that the use and purpose of the collected data be explicitly communicated, and that consent be actively and clearly granted by the user. Additionally, transparency requirements further imply that all processing purposes must be conveyed in a comprehensible manner [56]. However, the interpretation of what constitutes "comprehensible" is often subjective, and regulatory loopholes may exist [14][57].

Another key regulatory principle is accountability, which requires that all valid consent be demonstrably obtained and properly managed over time [56]. Consequently, consent must be treated as a system responsibility rather than a one-time interaction, with all consent records stored, versioned, and auditable [58]. Implementing this introduces challenges in data management, traceability, and scalability [59]. Additionally, GDPR establishes data subject rights, meaning that all collected personal data must be accessible, rectifiable, erasable, and subject to processing restrictions [60]. Therefore, consent architectures must support continuous updates while minimizing complexity.

2.6 Privacy by Design Principles

The Privacy by Design (PbD) framework was first developed by Ann Cavoukian as a "universal framework for the strongest protection of privacy in the modern era" [61]. The PbD framework comprises seven foundational principles and provides guidance for developing more detailed criteria for implementation, auditing, and verification. This approach has been incorporated into the General Data Protection Regulation (GDPR), and as a result, the two share many similarities in their underlying philosophy and practical objectives [62].

The first PbD principle states that privacy should be proactive rather than reactive [61]. In the context of consent systems, mechanisms should be designed from the outset to be non-manipulative and avoid dark patterns, reducing the need for corrective actions later [7]. The second PbD principle emphasizes that privacy should be the default, meaning users should not be required to take action to protect their privacy [61]. This principle can be operationalized through purpose specification, collection limitation, data minimization, and restrictions on use, retention, and disclosure. These measures align with GDPR regulations and dynamic consent principles and should be carefully considered when designing consent flows [55][11].

The third principle states that privacy must be embedded in design and architecture and should not be treated as a mere feature [61]. In consent management systems, functions such as permission handling, data storage, and user interaction mechanisms must be integral components of the system [63]. The fourth principle asserts that enforcing privacy should not compromise other system objectives. For consent flows, this means maintaining usability while adhering to privacy regulations [46].

The fifth PbD principle emphasizes end-to-end security, ensuring that data remains protected throughout its lifecycle [61]. Consent mechanisms should maintain accurate permission states over time, ensuring that revoked or modified consent is consistently respected across the system. The sixth principle focuses on transparency and visibility, meaning that data practices should be observable and auditable [61]. In consent flows, this supports the development of clear consent dashboards, understandable processing purposes, and accessible methods for users to modify their consent.

The final PbD principle highlights that privacy should be user-centered [61]. Systems should prioritize informed decision-making and user autonomy rather than optimizing solely for data collection or engagement metrics [64]. This aligns closely with user-centered consent flows, where interaction design aims to reduce cognitive burden, prevent consent fatigue, and enable meaningful control without overwhelming users.

Later work has revisited and refined the PbD framework from a more practical engineering perspective. Gürses et al. [65] argue that early interpretations of the framework remained relatively abstract and compliance-focused, making them less suitable for concrete system implementation. Their work places particular emphasis on data minimization, identifying it as a central engineering strategy for achieving PbD. They advocate strategies such as minimizing data collection, limiting disclosure, and restricting retention in order to enhance data privacy. From this perspective, effective consent mechanisms should operate within, and actively support, systems designed to reduce unnecessary data exposure.

2.7 Scalability Challenges in Consent Management

As digital services evolve into increasingly complex data ecosystems, consent management must operate across a growing number of actors, purposes, and data exchanges (see Section 2.1). In this context, scalability becomes a key challenge, particularly in ensuring that consent mechanisms remain understandable, meaningful, and manageable for users as the complexity of data exchanges increases.

A major contributor to this challenge is the growing number of actors involved in data processing, as user data is rarely accessible only to a single service provider. Instead, it is frequently shared with multiple third-party actors, such as advertisers, analytics providers, and external service integrations [66]. Each of these actors may process the data for different purposes. This increasing number of consent relationships makes it difficult for users to maintain a clear understanding of who has access to their personal data and for what purposes it is being used.

Increasing the granularity of consent options can enhance transparency and give users greater control over which data are used and by whom [67]. However, highly granular consent interfaces may also increase interaction complexity and reduce usability [68]. Similarly, providing greater

transparency regarding data processing practices strengthens accountability but can lead to information overload [69]. Designing consent flows therefore requires balancing transparency, simplicity, and user control in a way that remains manageable as the number of data interactions grows.

Some emerging approaches attempt to address these challenges through structured processes and models for user data exchanges. For example, decentralized data marketplaces such as Wibson [15] propose models that allow users to directly manage and authorize the processing of their data. These platforms demonstrate how structured data transactions can specify the purpose of data use, the parties involved, and the conditions under which data may be shared [15]. Although such decentralized architectures are not directly suitable or integrable with mainstream centralized digital service platforms [70], they offer valuable design principles, including explicit purpose specification, controlled data disclosure, and transparent transaction records. These ideas illustrate potential strategies that can be applied to enhance user control and transparency in consent management systems.

2.8 Web Application Development

Modern digital service platforms are typically implemented as web applications that operate using a client-server architecture [71]. In such systems, the client is responsible for rendering the user interface and handling interactions with the user, while data storage and back-end services are managed by a remote server. In modern web applications, this architecture has evolved into more client-heavy systems, where a larger portion of the application logic is executed on the client side [72]. These approaches are often implemented as Single Page Applications (SPAs), which dynamically update content without requiring a full page reload.

SPAs are commonly developed through JavaScript-based modern frameworks, which allows for a fluid and responsive user interfaces [72]. One widely adopted framework used in this context is React [73]. The React framework is component-based in which the user interface consists of reusable and independent components. Each component is implemented as a function with its own rendering logic and behavior, enabling modular development. This development approach is particular suitable in systems that involve multiple interactive views.

Managing application state in client-driven applications is a critical aspect of web development. State management mechanisms allow an application to maintain and coordinate information about user interactions, interface states, and application data across different components [74][75]. Consistent application state ensures that all user decisions are accurately reflected throughout the interface. Modern state management libraries, such as Zustand [76], provide client-side solutions with built-in persistence capabilities. This approach enables applications to store and retrieve states locally, for example through browser storage such as localStorage [77], without relying on external back-end services for data persistence [78]. As a result, architectural complexity can be reduced while maintaining responsiveness and ensuring data consistency within the client environment.

Architectural patterns are frequently employed in web development to structure user interfaces. One such pattern is the Model-View-Presenter architecture, which separates the application into three layers: application logic, user interface rendering, and interaction handling [79]. This separation improves maintainability and modularity in the development of interactive web applications.

2.9 Related Work

Emerging blockchain technologies have been proposed as a solution to give individuals greater control over their personal data. Wibson [15], a decentralized data marketplace that aims to enable individuals to monetize their personal information while maintaining anonymity through smart contracts and tokenization using Ethereum [80], represents one such implementation. The platform utilizes a flow of operation that guarantees that the data seller remains anonymous unless explicit consent for revealing their identity is given and ensures that more exact data than was requested is not leaked upon the data transfer. While Wibson addresses important privacy concerns, blockchain-based data marketplaces face substantial barriers to practical implementation. There are scalability issues as the transaction costs associated with smart contracts create economic friction [81] that undermines the platform's core value proposition, particularly for lower-value datasets.

The usability of consent interfaces has received increasing attention in privacy research, particularly regarding how design choices influence user decision-making and awareness. Beyond identifying problematic dark patterns in cookie consent interfaces [47], recent work has conducted comprehensive usability assessments to understand which design parameters actually impact user experience. A two-stage study combined heuristic inspection of consent interfaces with large-scale empirical testing [47]. It found that specific design choices, such as blocking functionality, inline cookie options, and persistent controls for changing consent decisions, significantly affect the extent to which users can make informed choices about their data [47]. Notably, consent interfaces that employ a fully-blocking design preventing users from proceeding without engaging with consent options, paired with inline cookie category selections and a persistent button for later consent changes, demonstrated superior usability outcomes compared to less restrictive alternatives [47]. These findings underscore that designing for informed consent requires careful attention to interaction patterns and interface layout, not merely the removal of manipulative design tactics. However, engagement with consent options does not automatically ensure informed consent.

While interface design improvements enhance usability, they do not automatically improve user comprehension of data implications. The finding that 71% of users misunderstand tracking [48] suggests that even well-designed consent flows risk enabling uninformed consent if they do not explicitly support user understanding of what data sharing entails. Critically, this research suggests that consent interfaces, even well-designed ones, must explicitly support user comprehension of data implications, particularly when incentives are introduced.

Chapter 3

Methods

This chapter presents the methodological approach used throughout the project. Section 3.1 describes the scientific methods applied when approaching the project work. Section 3.2 outlines the literature study conducted during the research phase and explains how it is used to derive the system requirements. Section 3.3 presents the agile development approach used during the design and development of the user-centered consent flow prototype. Finally, Section 3.4 describes the evaluation method used to assess the developed prototype.

3.1 Research Approach

This work is based on a qualitative research method [32] as the aim is to deliver a user-centered consent flow prototype grounded in research, rather than to test a specific hypothesis. More specifically, this project employs an applied research method. Applied research involves addressing a known and practical problem, often by leveraging existing research and real-world data to develop solutions and practical technologies [32]. This with the goal of contributing additional knowledge within the research field. In this case, an applied research method is appropriate, as the project aims to design and develop a practical, user-centered consent flow prototype.

3.2 Literature Review and Requirement Derivation

The collection of relevant theory and data that forms the basis for the derived system requirements is conducted through a structured literature study. Through this study, existing research and regulatory policies on digital service platforms, consent mechanisms, regulatory frameworks, and privacy-preserving system design are analyzed. The purpose of the literature study is to develop an understanding of the key insights and challenges related to consent mechanisms and management systems within modern data ecosystems, and to translate these insights into concrete system requirements for the prototype.

Based on the insights from the literature review, a set of system requirements is formulated to guide the design and development process of the consent flow prototype. The specified requirements are grounded in theoretical and legal principles and are translated into practical design objectives that the consent flow should support. Software system requirements are usually, and in this case as well, classified as either functional or non-functional [82]. Functional requirements describe what the system shall do. For instance, what services it should provide, how it should react to particular inputs, and how the system should behave in certain situations. Functional requirements may also include what a system should not do. In contrast, non-functional requirements describe the constraints on the functions or services offered by the system. These requirements often refer to the system as a whole rather than to individual system features. The identified requirements act as the backbone of how the prototype is designed, while also serving as criteria later in the evaluation process.

3.3 System Design and Development

During the design and development process of the user-centered consent flow prototype, an Agile-inspired workflow is applied [83]. The Agile Manifesto identifies a set of values and principles intended to guide the software development process [84]. Agile methodologies emphasize iterative and incremental development, allowing system components to be implemented and refined progressively. The purpose of applying this method in the project is to support an organized development process while enabling continuous refinement of implemented features throughout the prototype development.

In this approach, development tasks are organized in a product backlog and implemented incrementally. This allows development to progress through smaller iterations where individual components can be evaluated and adjusted as the system evolves. Compared to traditional sequential development approaches such as the waterfall model, where development stages are typically completed in a fixed order, the iterative nature of Agile development enables easier integration and evaluation of system components throughout the process.

To support this workflow, a Kanban board [85] is used to visualize and manage development tasks. This tool provides an overview of the development workflow by organizing tasks into manageable units and displaying them across different stages of progress. In practice, tasks are categorized into stages representing planned tasks, ongoing work, and completed tasks, allowing the development progress to be monitored throughout the project. Additionally, the Kanban method limits the number of tasks that can be in progress simultaneously, which helps maintain focus and reduces workflow bottlenecks during development. By structuring tasks in this way, the development process remains organized while supporting incremental implementation of the consent flow prototype.

3.4 Evaluation Method

To assess whether the developed consent flow prototype fulfills the objectives of the study, a criteria-based evaluation method is applied [33]. The purpose of the evaluation is to determine how well the prototype meets the identified requirements derived from the literature study. Therefore, the evaluation criteria in this case are the aforementioned derived requirements, focusing on aspects such as user control, privacy, scalability, and transparency. By evaluating the prototype against these requirements, it becomes possible to determine whether the system successfully translates theoretical principles into practical system functionality.

The evaluation is conducted through a systematic analysis of the prototype against the derived requirements. For functional requirements, each requirement serves as a measurable criterion, and the evaluation applies a binary scoring method to determine whether the prototype fulfills the requirement or not. In this method, each requirement is assessed as either fulfilled (1) or not fulfilled (0). These requirements correspond to concrete system functionalities, such as specific features or interactions within the consent flow. Therefore, they can be objectively assessed by examining whether the implemented system behavior aligns with the intended functionality. This approach provides a reproducible and transparent evaluation process.

Non-functional requirements concern qualitative aspects such as usability, perceived privacy, and transparency, which are not easily measurable through system inspection alone. To complement this analysis, user testing is conducted using a convenience sampling [86] approach, where participants interact with the prototype and provide feedback. While this method has inherent drawbacks, such as bias, self-selection, and limited representativeness, it offers practical advantages in this context: participants are readily accessible, data collection is less time-consuming, and it provides qualitative insights from real user experiences that help contextualize subjective assessments [87]. Although findings from this convenience sample cannot be generalized to broader populations, the approach is suitable for this project.

The developed questionnaire consists of statements to which participants indicate their level of agreement. A Likert scale [88], a common response format in user experience research, is employed. The scale provides a clear and intuitive way to express agreement across five levels. These range from 1 (strongly disagree) to 5 (strongly agree), with intermediate options of 2 (disagree), 3 (neutral), and 4 (agree). Each statement addresses a non-functional requirement category. Open-ended questions follow, inviting detailed feedback. The full questionnaire is provided in Appendix A. This approach enables both quantifiable assessment and nuanced contextual feedback.

Chapter 4

Design Criteria

This chapter presents the pre-study and the derived requirements that form the foundation of the developed prototype. Section 4.1 outlines the key insights from the literature study, while Section 4.2 and Section 4.3 describes the derived functional and non-functional requirements of the system.

4.1 Key Insights from Literature

The literature review highlights several key insights and considerations that directly influence the design and shape the requirements of the user-centered consent flow prototype.

First, digital service platforms operate in complex digital environments. Within these environments, multiple actors, purposes, and data exchanges are involved. This increasing complexity makes it more difficult for users to maintain an understanding of how their data is being processed. Therefore, there is an increasing need for more transparent and structured consent mechanisms that clearly communicate this information.

Second, existing consent mechanisms often fail to provide meaningful user control, even though regulatory frameworks such as the General Data Protection Regulation (GDPR) attempt to enforce strict requirements for valid consent. This is because practical implementations often rely on complex interfaces, extensive privacy policies, and manipulative design patterns when requesting consent. These so-called dark patterns can influence users to give consent without fully understanding their choices. This results in consent that is formally valid but lacks practical meaning.

Third, the concept of dynamic consent introduces an alternative approach to managing consent that provides users with greater control. Dynamic consent treats consent as an ongoing life-cycle rather than a one-time event, allowing users to modify or revoke their consent over time. This highlights the need for systems to support continuous interactions, including mechanisms for reviewing, modifying, and revoking consent.

Although increased control over consent options may be beneficial, empirical research shows that the usability of consent mechanisms must also be carefully considered. Users demonstrate a preference for simplified, slider-based designs over complex alternatives, yet they show significant misunderstandings of consent implications and susceptibility to default option influence. Granular consent options provide detailed control over data sharing and contribute to increased transparency and user autonomy. However, excessive granularity may increase cognitive burden and contribute to consent fatigue. Therefore, a balance between user control, transparency, and usability must be achieved when designing consent flows.

In addition, regulatory frameworks and privacy principles, such as the GDPR and Ann Cavoukian's Privacy by Design (PbD) principles, impose requirements including transparency, accountability, and data minimization. These ensure that consent is properly managed, documented, and enforced throughout its life-cycle. Such requirements highlight the importance of both effective user interaction and back-end consistency within consent mechanisms.

Scalability in consent management is a key challenge when designing user-centered systems. As the number of data interactions, third-party actors, and processing purposes grows, consent mechanisms must remain understandable and manageable without overwhelming users. While increasing granularity can enhance transparency and user control, it simultaneously increases interaction complexity and risks information overload. Designing consent flows therefore requires balancing transparency, simplicity, and user control in a way that remains effective as data ecosystems grow in complexity. Emerging approaches, such as decentralized data marketplaces, offer design principles (explicit purpose specification, controlled data disclosure, and transparent transaction records) that may inform more scalable consent mechanisms.

From a practical engineering perspective, PbD should be operationalized through concrete data minimization strategies (reducing collection scope, limiting disclosure, and restricting retention periods) rather than being treated as abstract principles. This approach ensures that consent systems operate within a broader architecture designed to minimize unnecessary data exposure.

Empirical research on consent interface usability has identified specific design patterns that enhance user decision-making. Interfaces employing fully-blocking designs (preventing bypass without engagement), paired with inline consent options and persistent controls for later modification, demonstrate superior usability outcomes. However, a critical limitation is that improved interface design alone does not guarantee meaningful user comprehension or control. Even well-designed consent flows risk enabling uninformed compliance if they do not explicitly support users understanding of data implications or operate within systems that minimize unnecessary data collection. This suggests that truly effective consent mechanisms require a holistic approach: combining sound interface design with explicit comprehension support, transparent purpose specification, and backend architectures designed to minimize data exposure by default.

4.2 Functional System Requirements

The functional requirements are derived from the conducted literature review and define the specific actions and behaviors the system must support to enable a user-centered consent flow. They describe what the system should do and provide a concrete framework for the development of the prototype. A compilation of the derived functional requirements are presented in Table 4.1.

ID	Name	Description
FR1	Transparent Consent Request Presentation	Shall clearly convey what personal data is being collected, the purpose of processing, and which parties are involved.
FR2	Explicit Consent Action	Shall actively require consent through an explicit action, without reliance on preselected options.
FR3	Granular Consent Selection	Shall allow users to grant or reject consent for individual data processing purposes or categories where possible.
FR4	Consent Modification	Shall allow users to modify the state of previously granted consent.
FR5	Consent Withdrawal	Shall provide an accessible interface that allows users to revoke previously granted consent.
FR6	Consent Overview Interface	Shall provide a dashboard or overview interface where users can view all their current consent decisions.
FR7	Third-party Visibility	Shall display the third-party actors involved in data processing when possible.
FR8	Purpose Transparency	Shall communicate the intended purpose of data processing in a concise and understandable manner.
FR9	Consent Record Storage	Shall store consent decisions and maintain records of user decisions.
FR10	Data Access Control Enforcement	Shall enforce current consent states in all data sharing actions.

Table 4.1: Functional Requirements

The first functional requirement, FR1, specifies a transparent consent request interface. In this context, transparency means clearly presenting the personal data being collected, the purpose of processing, and any involved third parties. This requirement is based on the General Data Protection Regulation's (GDPR's) definition of informed and specific consent, as discussed in Section 2.2 and Section 2.5.

The second functional requirement, FR2, specifies that all consent must be obtained through an explicit user action, without relying on preselected options. This is based on the GDPR's requirement that consent must be freely given and unambiguous, as discussed in Section 2.5.

The third functional requirement, FR3, specifies granular consent options, allowing users to selectively consent to specific data processing purposes or categories where possible. This requirement is grounded in granular consent design, as discussed in Section 2.4.

The fourth functional requirement, FR4, specifies the user's ability to modify previously granted consent. This may include changing eligibility or adjusting more fine-grained preferences, such as data processing purposes. This requirement is based on the concept of dynamic consent as an ongoing lifecycle, discussed in Section 2.4.

The fifth functional requirement, FR5, specifies that users must be able to withdraw previously granted consent through an easily accessible interface. This is based on the GDPR's requirement that consent must be revocable, as discussed in Section 2.2 and Section 2.5.

The sixth functional requirement, FR6, specifies a consent overview interface that provides a summary of all current consent decisions, for example in the form of a dashboard. This requirement is based on transparency and user awareness of data-sharing relationships, as discussed in Section 2.2 and Section 2.5.

The seventh functional requirement, FR7, specifies third-party visibility, meaning that all third-party actors involved in data processing should be clearly presented whenever possible. This requirement is based on multiparty data sharing, as discussed in Section 2.1 and Section 2.2.

The eighth functional requirement, FR8, specifies transparency of data processing purposes, which should be communicated in a concise and understandable manner. This requirement is based on the GDPR's requirement and the Privacy by Design (PbD) principle of purpose transparency, as discussed in Section 2.2 and Section 2.6.

The ninth functional requirement, FR9, specifies maintaining a record of consent decisions. This requirement is based on the GDPR's accountability principle, as discussed in Section 2.5.

Finally, the tenth functional requirement, FR10, specifies the enforcement of consent states and decisions in all data-sharing actions. This is based on the PbD end-to-end security principle, as discussed in Section 2.6.

4.3 Non-functional System Requirements

Non-functional requirements are likewise derived from the conducted literature review but focus on the system's qualities and characteristics rather than its specific actions and behaviors. They specify how the system should perform and outline constraints and attributes necessary to support an effective user-centered consent flow. A compilation of the derived non-functional requirements is presented in Table 4.2.

ID	Name	Description
NFR1	Transparency	Consent interfaces should communicate data processing practices in a clear, structured, and understandable manner.
NFR2	Usability	Consent flow should minimize cognitive effort and avoid overwhelming users, supporting comprehension and manageable decision-making.
NFR3	Avoidance of Dark Patterns	Interface should avoid manipulative design techniques such as preselected options, misleading elements, and hidden privacy controls.
NFR4	User Control	Users should easily be able to view consent decisions, modify permissions, and withdraw consent.
NFR5	Privacy by Default	The system should minimize data exposure by default and require explicit user permission before any data sharing.
NFR6	Scalability of Consent Management	The consent flow should remain manageable and understandable as the number of data categories, processing purposes, and third-party actors increases.
NFR7	Engagement	Data requests should be presented in a way that facilitates active user engagement and informed decision-making.

Table 4.2: Non-functional Requirements

The first non-functional requirement, NFR1, specifies that the system should be transparent. In this context, all consent-related interfaces must communicate data processing activities in a clear, structured, and understandable manner. This requirement is based on the General Data Protection Regulation's (GDPR's) and Privacy by Design (PbD) transparency principles discussed in Section 2.5 and Section 2.6.

The second non-functional requirement, NFR2, specifies that the system should promote usability. More specifically, the consent flow should minimize cognitive effort and avoid overwhelming users, thereby supporting user comprehension and decision-making. This requirement is based on consent usability considerations discussed in Section 2.2.

The third non-functional requirement, NFR3, specifies the avoidance of dark patterns in the system. This includes practices such as preselected options, visually misleading interface elements, and hidden privacy controls. This requirement is based on discussions of dark patterns in consent interfaces in Section 2.2, Section 2.6, and Section 2.9.

The fourth non-functional requirement, NFR4, specifies that the system should promote user control. This includes making consent decisions easily viewable, as well as enabling accessible modification and withdrawal of consent. This requirement is central to the research question of the report.

The fifth non-functional requirement, NFR5, specifies that the system should support privacy by default. This means that personal data exposure should be minimal by default, and any additional data sharing should require explicit user permission. This requirement is based on the PbD principle of privacy by default, as discussed in Section 2.6.

The sixth non-functional requirement, NFR6, specifies scalable consent management. The consent flow should remain manageable and understandable as the number of data categories, processing purposes, and third-party actors increases. This requirement addresses the scalability challenges discussed in Section 2.7.

Finally, the seventh non-functional requirement, NFR7, specifies that the system should support active user engagement. This means that data requests should be presented in a way that facilitates engagement and informed decision-making. This requirement is based on research discussed in Section 2.3.

Chapter 5

Prototype Design and Implementation

This chapter presents the design and implementation of the developed user-centered consent flow prototype. Section 5.1 provides an overview of the system, including its purpose and limitations. Section 5.2 describes the overall architecture of the prototype. Section 5.3 outlines the modeling of consent and its lifecycle within the system. Finally, Section 5.4 presents the design, interfaces, and functionalities of the system.

5.1 System Overview

The prototype is implemented as a web application and focuses primarily on the design of interaction flows, consent management, and related functionalities, rather than serving as a complete production-ready system. In line with the delimitation of the project, the implemented system operates within a simplified environment using mock data and does not include a full-scale back-end infrastructure or advanced security mechanisms. Instead, emphasis is placed on how consent flows can be designed to support key principles such as transparency, user control, and usability.

The prototype includes several core components that reflect the lifecycle of consent management. Users can interact with service offerings and data offers, provide consent through structured consent interfaces, and manage their existing consent via a dedicated dashboard. In addition, the system includes mechanisms to track consent-related actions, enabling visibility into how consent is granted and modified over time.

Overall, this prototype serves as a proof-of-concept implementation that translates theoretical principles from the literature into practical system functionalities. The development process is guided by the derived requirements, which provide a framework for the design and implementation of the system. In this way, the prototype demonstrates how user-centered consent flows can be designed and potentially integrated into existing digital service platforms, while remaining understandable and manageable from the user's perspective.

5.2 System Architecture

Figure 5.1 illustrates the overall architecture of the prototype, showcasing the interaction between the presentation layer, state management, and client-side persistence. The system follows a layered structure in which data flows bidirectionally between components, enabling real time updates and state rehydration upon application upload.

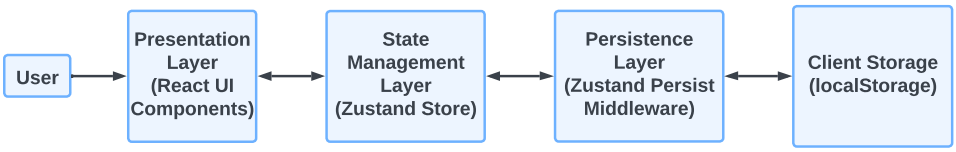


Figure 5.1: User to client-side architecture diagram.

The prototype is implemented as a client-driven web application, following a Single Page Application (SPA) architecture facilitated by the React framework. This approach allows all of the application logic to be executed on the client side, enabling seamless transitions between different interface views without the need for full page reloads. This architectural choice supports the development of interactive consent flows and allows user actions to be reflected directly within the interface.

The user interface is built using a component-based architecture, where the application is divided into modular and reusable components. Each component is responsible for rendering different parts of the interface. In addition, the application is developed following the Model-View-Presenter architecture, clearly separating each interface view into corresponding presenter and view components. This modular structure supports the management of the different parts of the consent flow and facilitates the incremental development of the prototype.

To maintain consistency throughout the prototype, a centralized state management approach is employed using Zustand. The application state stores information related to user consent, service offerings, and interaction history. This enables different components of the system to access and update shared data, ensuring that all information, including user decisions, is consistently reflected throughout the interface.

For data persistence, the prototype utilizes a client-side storage mechanism implemented through Zustand's persistence middleware. This approach allows all consent-related data to be stored locally in the user's browser using localStorage. As a result, user data persists across sessions and page reloads without requiring a dedicated back-end service or external database. For the purpose of a lightweight prototype implementation, this approach offers sufficient functionality without unnecessary complexity.

5.3 Consent Model and Lifecycle

In the prototype, consent is modeled as a structured and dynamic object that represents the relationship between a user, a data request, and the conditions under which personal data may be processed. The system adopts a dynamic perspective, where consent evolves over time rather than being treated as a one-time event. This reflects the principles of dynamic consent discussed in Section 2.4, allowing users to modify and update their decisions continuously.

The consent model is implemented as a JavaScript data structure containing essential attributes required to support transparency, user control, and traceability, as illustrated in Figure 5.2. The `serviceId` identifies the service requesting data, while `purposes` allows for fine-grained consent decisions. Each entry includes a `granted` boolean, enabling users to approve or deny specific purposes and the corresponding data types. The `thirdParties` field contains all third-party sharing of the approved user data, which can be easily displayed to the user. The `status` field reflects the current state of consent, which may be `active`, `expired`, or `revoked`. The `timestamps` provide a clear record of when consent was created, updated, expired, or withdrawn. A `duration` field tracks the period a consent remains active before it expires and requires renewal. Additional `metadata`, such as the `source` of the consent request, supports contextual understanding.

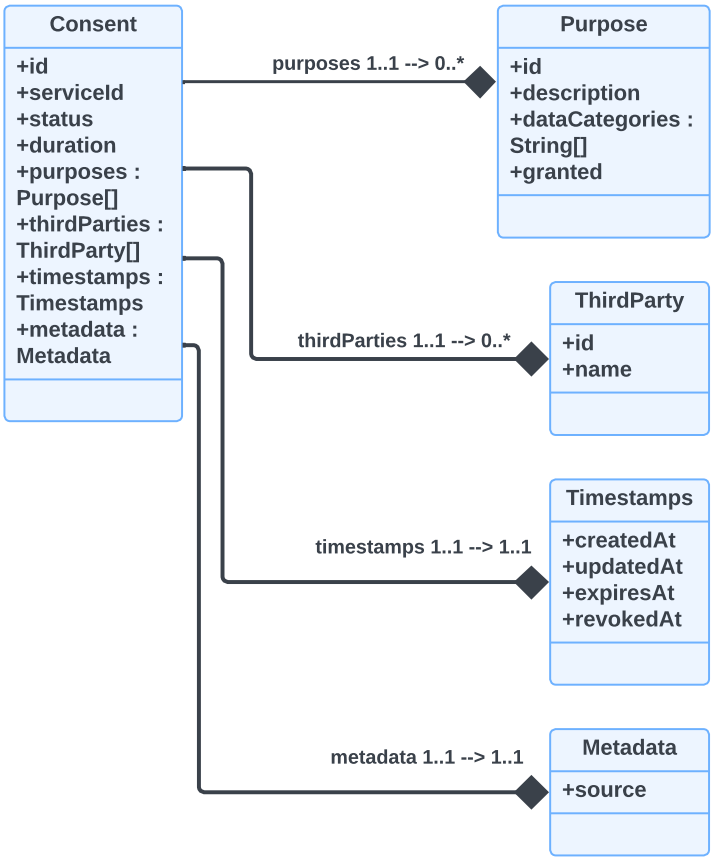


Figure 5.2: JavaScript implementation of the consent object

The lifecycle of consent within the system is defined through a set of states and transitions, illustrated in Figure 5.3. Initially, consent is either granted or declined by the user through a structured interface, entering the corresponding state. If active consent exists, the user may modify its scope, resulting in an updated state. Conversely, the user may withdraw consent entirely, resulting in a revoked state. These transitions are triggered by user interactions and are immediately reflected in the system state. Consent may also expire automatically, which is triggered by the system.

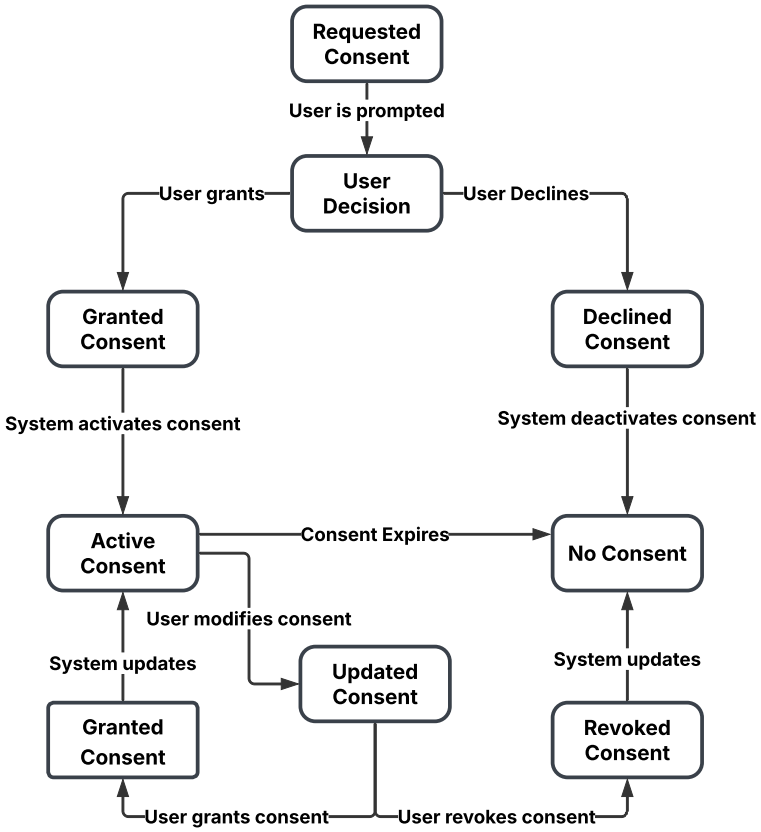


Figure 5.3: Consent lifecycle state diagram

To support this lifecycle, the system ensures that consent-related changes are consistently propagated throughout all components. When a user modifies or revokes consent, or when consent expires, the updated state is reflected across relevant interfaces, such as service interactions and the consent dashboard. This ensures that user decisions are respected in real time and that the system maintains an accurate representation of current permissions.

5.4 Consent Flow Design

The design of the consent flow prototype, available at [89] is centered on structuring user interactions in a way that supports transparency, meaningful user control, and usability within a complex data environment. Rather than treating consent as a one-time interaction, the system organizes consent across different states in its lifecycle. The interaction flow is therefore designed to accommodate management across these stages. In addition, the consent flow is designed in a way that could potentially be integrated into modern digital service platforms, either partially or more comprehensively. The flow of the prototype across its different interface views is illustrated in Figure 5.4.

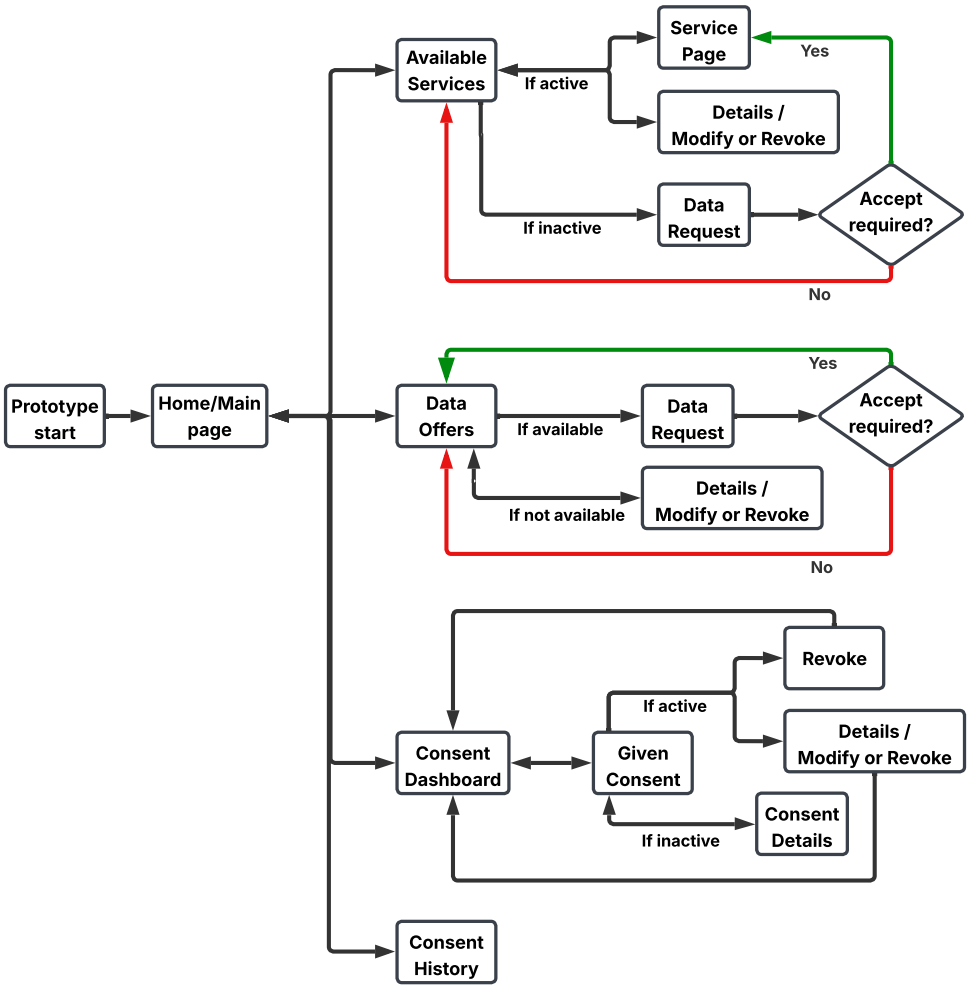


Figure 5.4: Flow diagram of the consent flow prototype

The user starts at the system's main page, which provides access to four primary interface views: available services, data offers, the consent dashboard, and the consent history. This structure separates different user intentions. Accessing either a service or a data offer represents forward interactions, where an incentive is provided in exchange for user data. In contrast, the consent dashboard and consent history support reflective interaction, allowing users to review and manage previously given consent.

When navigating to the available services view, as shown in Figure 5.5, users are presented with a list of services that may require access to personal data. This view represents one part of the system's functionality, where users can explore and access different services based on their needs. In a real world digital service platform, such interactions could occur when users select features such as booking services, connecting third party integrations, or enabling personalized content. From the user's perspective, this view serves as a space where services can be selected, with potential data sharing tied to those choices.

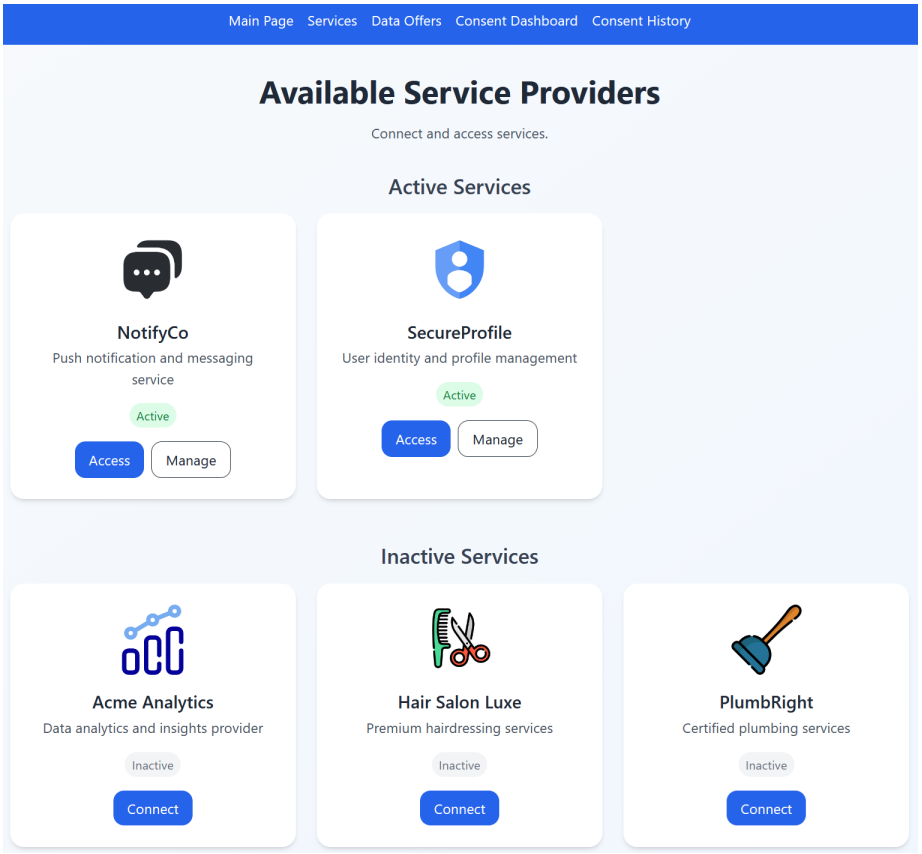


Figure 5.5: Available services view

The prototype is a generalized service environment in which multiple services are available. The same principles apply to platforms offering a single service, where data requests are still triggered at the point of use. In both cases, the prototype design ensures that consent is contextualized within the intended action of the user, rather than being presented in an isolated or abstract manner.

If a service does not require additional consent, it can be accessed directly. However, if the service requires data access, the user is guided to a structured data request interface, as shown in Figure 5.6. This interface presents information about the requested data, including purposes, corresponding data categories, and involved third parties.

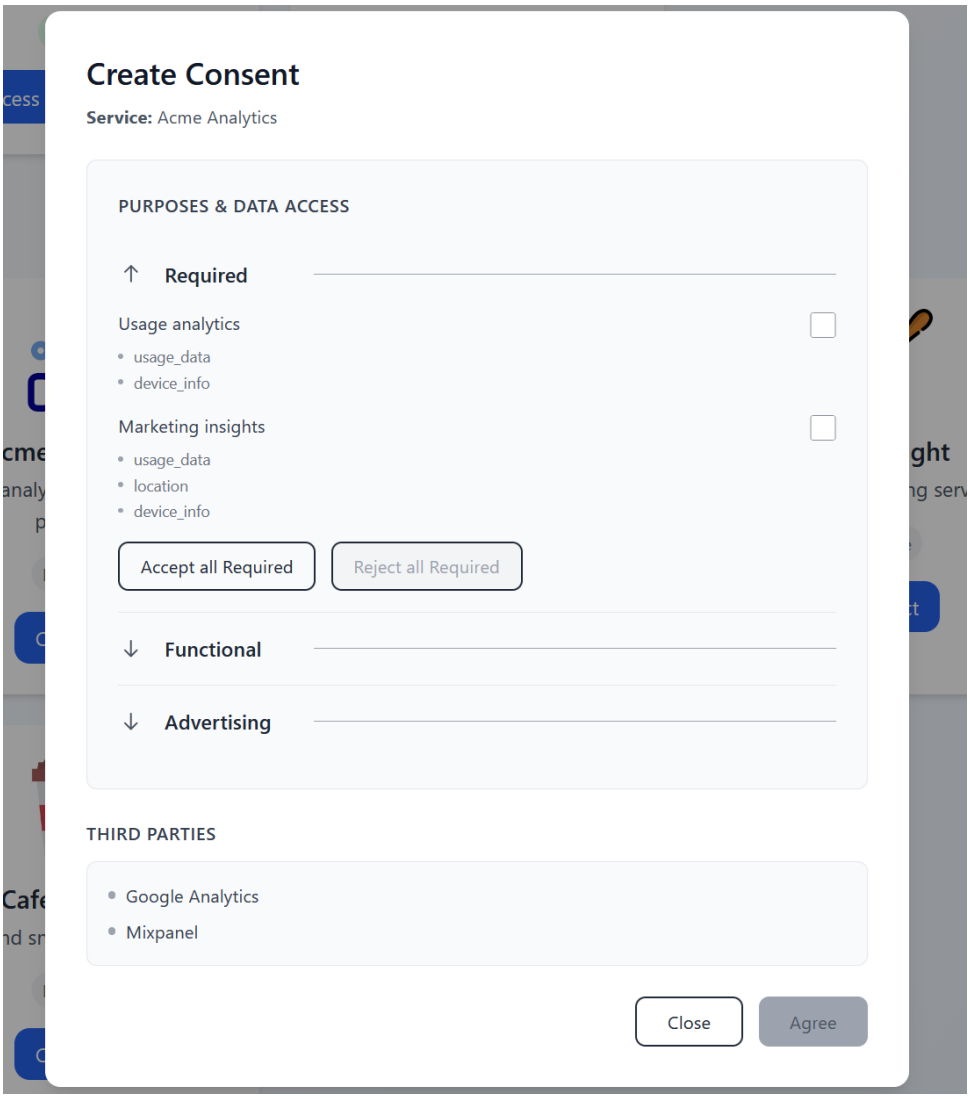


Figure 5.6: Data request view

The purposes in the data request interface are divided into three categories: necessary, functional, and advertising, to support clearer cognitive interpretation of the information. To enhance transparency and user control, dropdown menus are implemented within each category, allowing users to inspect more detailed information and adjust granular consent options before making a decision.

While research on consent interface design indicates that slider-based designs are generally preferred by users and that gesture-based interactions such as drag-and-drop and swiping can increase user attentiveness compared to traditional checkboxes, both approaches present limitations for this prototype. Slider designs offer insufficient granularity for fine-grained consent control across multiple purposes, while gesture-based interactions present scalability challenges when managing many data purposes. To address these constraints, the prototype employs checkboxes for individual purpose selection, paired with category-level options (”Accept All” and ”Reject All” buttons) to support usability without sacrificing control. Additionally, maintaining a unified interaction pattern across both the services and data offers contexts reduces cognitive load and interaction learning burden, prioritizing consistency over optimizing each section independently.

The decision-making step is designed to clearly reflect the consequences of user choices. If the user accepts the request, consent is granted and the service becomes accessible. If the user declines, the outcome depends on whether the requested data is essential for the service. In cases where some data is not strictly required, the user may still access the service, supporting voluntary participation. In contrast, if the data is required, access is denied and the user is returned to the available services view. This distinction clarifies the relationship between functionality and data usage, thereby improving transparency.

In the data offers view, as shown in Figure 5.7, users are presented with requests to share their data in exchange for a reward. When accepting one of these offers, the user is presented with the same data request interface as in the available services view. Each request includes a duration parameter that specifies the time period during which the user's data is shared. During this period, users retain the ability to modify or withdraw consent, potentially resulting in a reduction or loss of the associated reward.

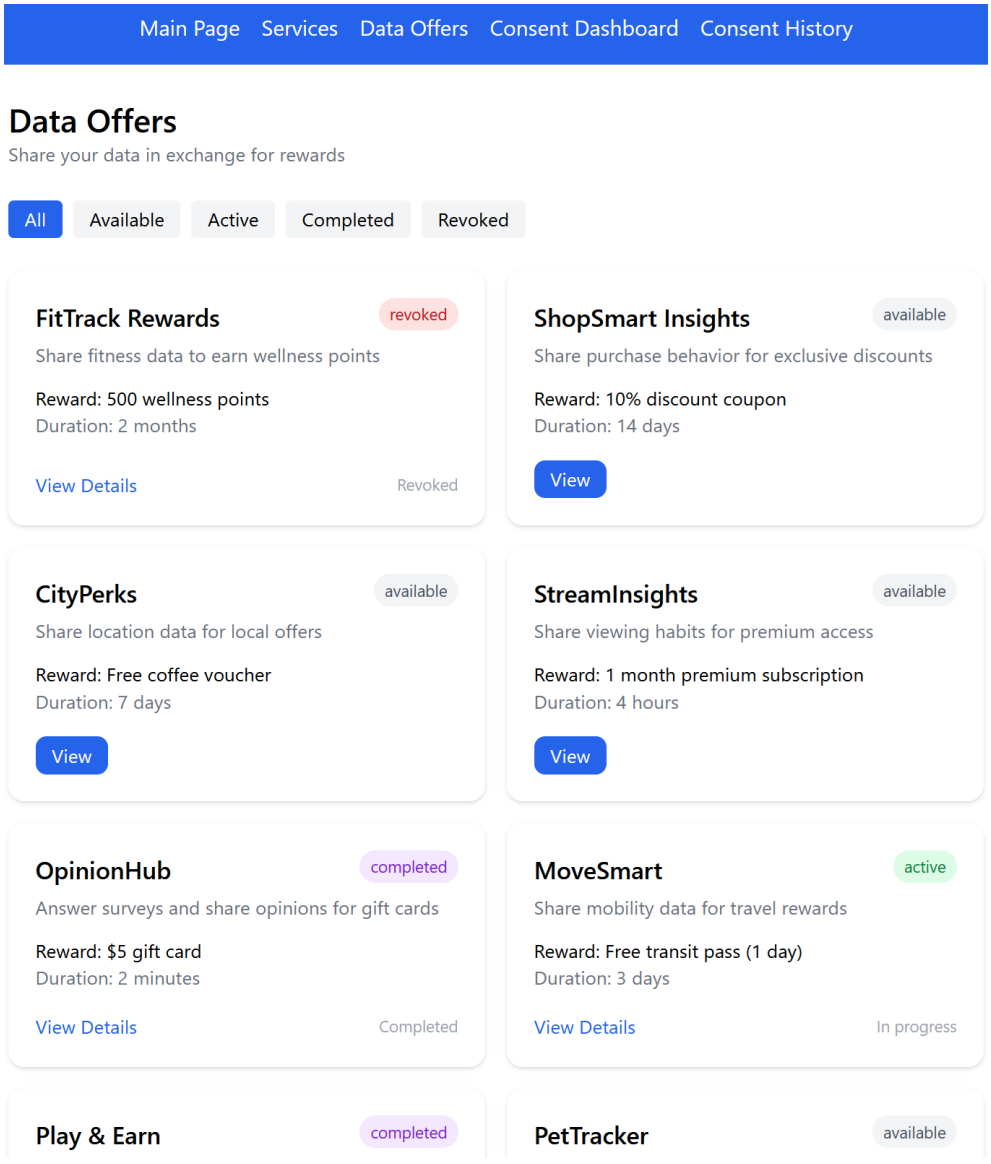


Figure 5.7: Data offers view

In addition to initial consent decisions, the system provides a dedicated interface for managing existing consent. Through the consent dashboard, as shown in Figure 5.8, users can view all previously granted permissions and access detailed information about each consent instance. From this view, users can modify or revoke active consent, reflecting the dynamic nature of consent management within the system. These interactions are directly connected to the consent lifecycle, ensuring that changes are consistently applied throughout the system.

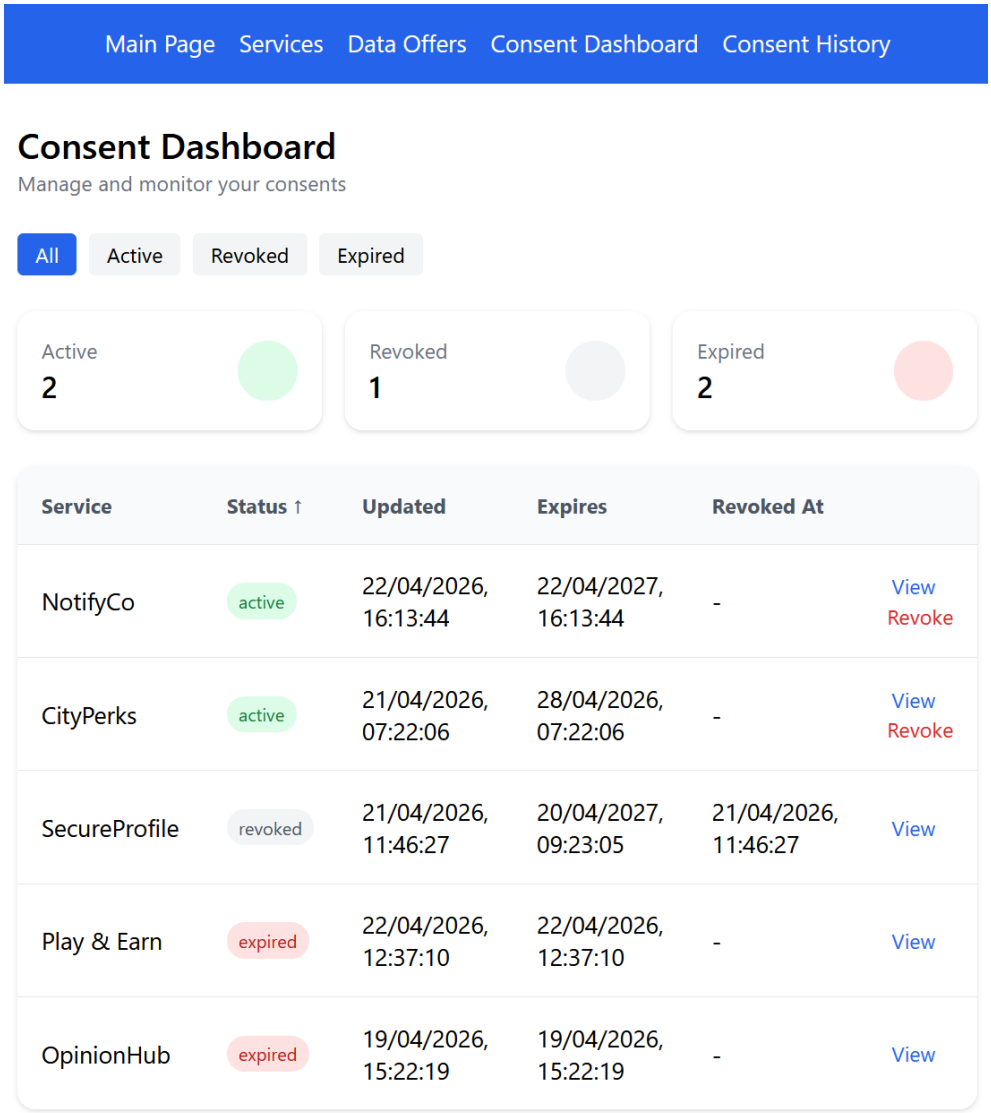


Figure 5.8: Consent dashboard

The consent history view, as shown in Figure 5.9, complements this functionality by providing a record of all past consent interactions, including granted, revoked, expired, or modified consent, whether initiated by the user or the system. This supports visibility and accountability by allowing users to trace how their data permissions have evolved over time. It also enhances perceived user control by providing easy access to detailed information whenever changes occur.

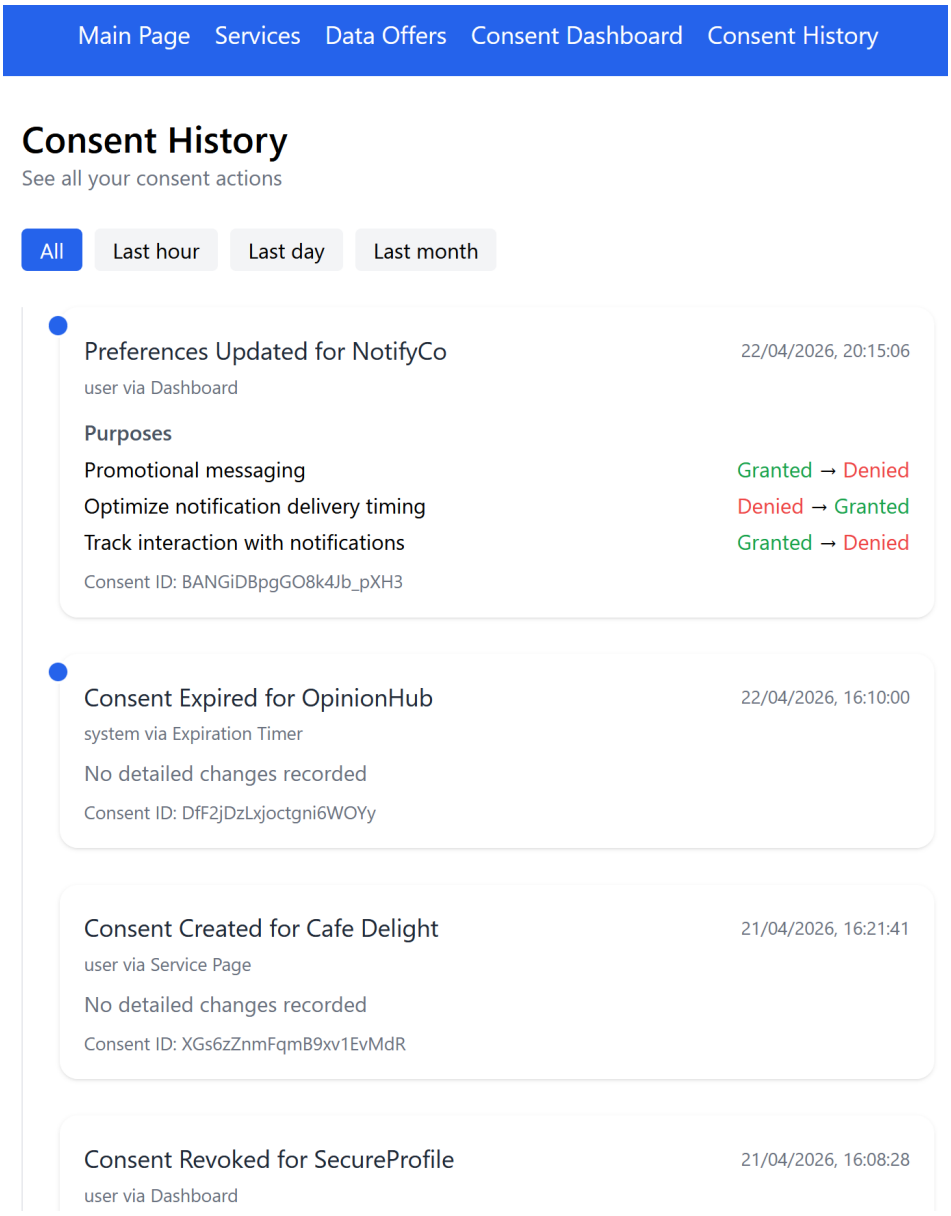


Figure 5.9: Consent history view

Overall, the system flow is designed to balance granularity with usability. While users are given control over detailed consent options, these are presented progressively through layered interfaces rather than all at once. This reduces cognitive overload and helps mitigate consent fatigue, while still enabling informed decision-making. By structuring interactions across distinct but connected interface views, the prototype aims to provide a scalable approach to consent management that remains understandable as the number of data interactions increases.

Chapter 6

Requirements Evaluation

This chapter outlines the evaluation process and presents the results of both the requirement evaluation and user testing of the prototype. Section 6.1 details the user testing procedure and its results. Section 6.2 presents the evaluation of the functional requirements, while Section 6.3 covers the evaluation of the non-functional requirements. Finally, Section 6.4 discusses the reliability and validity of the results.

6.1 User Testing

Through convenience sampling, fifteen participants were recruited. The selection criteria emphasized diversity in gender, age, and technical competence, as this was considered most suitable for approximating an average user perspective. During user testing, fifteen survey responses were collected. Results are analyzed based on the proportion of positive responses (rated 4 or higher on the Likert scale), corresponding to an ”agree” or ”strongly agree” answer. The responses are grouped by non-functional requirements, with each statement corresponding to a specific requirement category. The categories transparency, usability, avoidance of dark patterns, and user control are each represented by two statements, while the remaining categories are represented by a single statement. The grouping of statements per non-functional requirement is presented in Appendix A. Figure 6.1 illustrates the percentage of positive responses for each category.

Participants generally rated the system positively across most categories. Avoidance of Dark Patterns received the highest proportion of positive responses (97%), followed by Usability, User Control, Privacy by Default, and Engagement (each 87%). Transparency achieved 73%, while Scalability received the lowest proportion at 53%, indicating potential challenges when the consent flow is extended to complex scenarios. These patterns are further analyzed in Section 6.3 against the specific non-functional requirements.

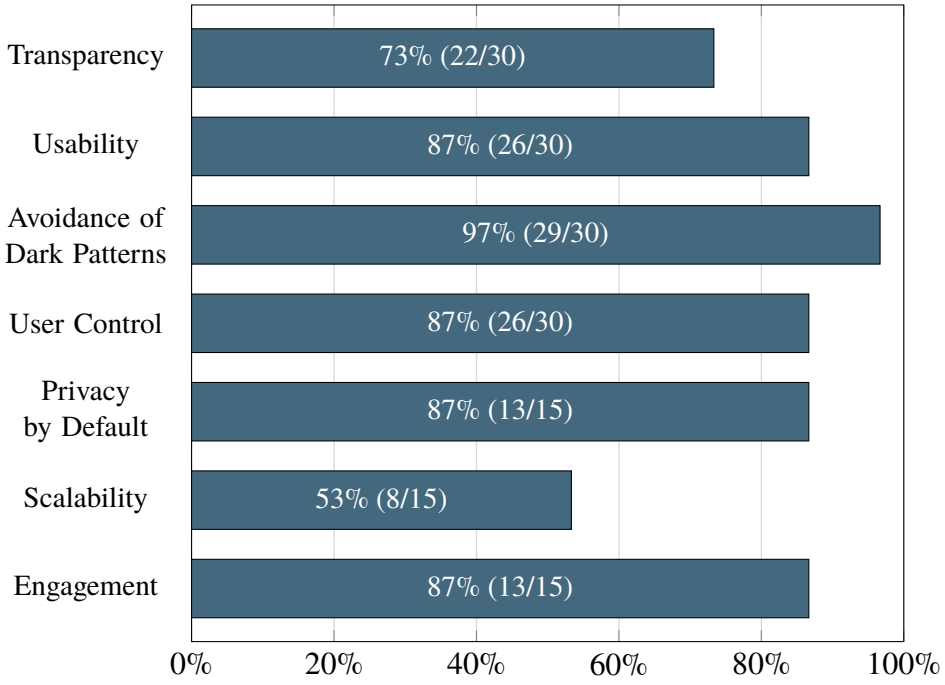


Figure 6.1: Percentage of responses rated agree or higher (≥4) by category

In addition to Likert scale responses, participants provided open-ended feedback on perceived strengths, weaknesses, and areas for improvement. Analysis of this feedback reveals several key themes.

Dashboard and Consent Management: The consent dashboard and history features received consistent positive feedback, with multiple participants identifying these as the most helpful aspects of the system. Participants appreciated the ability to view and revoke consent easily in a central location, with one noting that the dashboard ”would be massively helpful with additional graphs and stats over different types of consent/information.”

Information Density and Cognitive Load: Several participants reported experiencing cognitive overload when interacting with certain interface elements. Feedback included observations that the system ”feels slightly overwhelming when opening multiple dropdowns” and that ”too much detail in dropdowns” and ”too many categories when accepting consent” created challenges. One participant explicitly noted that the ”visual design/text interface is a bit cluttered and feels information dense at times, which might be especially cumbersome for people with dyslexia or similar disabilities.” This theme aligns with the lower Scalability rating (53%) and suggests that while the system functions clearly at baseline complexity, maintaining usability becomes more difficult as information volume increases.

Contextual Clarity and Explanations: Participants identified gaps in understanding the reasoning and implications of consent decisions. Feedback indicated that ”distinctions between purposes could be clearer” and that ”duration in data offers could be clearer”. Additionally, some participants noted confusion about specific functionalities, such as the distinction between required and optional consent options.

Suggested Improvements: Participants proposed several enhancements, including the use of visual summaries alongside textual information, plain language explanations and tutorials, progressive disclosure to reduce information density, and additional features such as consent expiration notifications.

6.2 Functional Requirements Evaluation

The functional requirements specified in Section 4.2 define the expected functionalities of a user-centered consent flow. Overall, the prototype satisfies these requirements, however certain limitations remain due to the scope of the implementation and its nature as a proof of concept.

Requirement FR1 specifies that the system should clearly convey the data collected, the purpose of processing, and all involved parties when requesting consent. This is achieved by categorizing purposes into three distinct groups: necessary, functional, and advertising. Within these categories, the relevant purposes are clearly listed along with the corresponding data categories involved. In addition, both the main service and any third parties are explicitly displayed within the interface. While this structure improves clarity, the effectiveness of the communication ultimately depends on the user’s ability to interpret the provided information, which may vary depending on their level of technical understanding.

Requirement FR2 specifies that all consent must be obtained through explicit user action, without reliance on preselected options. Within the system, consent is only granted when the user actively agrees to the processing purposes and explicitly clicks the “Agree” button. All processing purposes must also be actively selected through default unchecked options or via the “Accept all” button. This requirement is therefore satisfied, however the inclusion of an “Accept all” option, while practical, could potentially encourage less informed consent.

Requirement FR3 specifies that users should be able to granularly grant or reject individual processing purposes where possible. The prototype implements a dropdown menu within the broader purpose categories, allowing each purpose to be selected individually. In addition, within the consent dashboard, all active consent can be modified, where non-essential purposes can be granted or denied. Although the dropdown design helps mitigate complexity, the availability of fine-grained options may still lead to cognitive overload or decision fatigue for some users.

Requirements FR4 and FR5 specify that an accessible interface should be provided where users can easily modify or revoke consent. This is achieved through the consent dashboard, where all active consent can be modified or revoked. Although the functionality is available and accessible, its effectiveness depends on user awareness and willingness to actively use the dashboard.

Requirement FR6 specifies that the system should provide an interface where all given consent and related decisions are easily accessible. This is achieved through the consent dashboard and the consent history page, where all consent instances and related actions can be viewed. While this supports transparency, the presentation of historical data may become difficult to navigate as the number of consent interactions increases, indicating a potential scalability limitation. This is partially mitigated through filtering options, allowing consent decisions to be organized by time.

Requirement FR7 specifies that the system should display all third-party actors involved in data processing where applicable. This is implemented both in the consent request interface and when viewing specific consent instances in the dashboard. However, while third parties are listed, the prototype does not provide deeper insight into their roles or data handling practices, which may limit the user’s ability to make fully informed decisions.

Requirement FR8 specifies that all data processing should clearly communicate its intended purpose in a concise and understandable manner. This is achieved by presenting processing purposes alongside their related data categories, making it clear which data is used for each purpose. Although this improves clarity, the descriptions remain relatively high-level, and further refinement or standardization of purpose descriptions could enhance user understanding.

Requirement FR9 specifies that the system shall maintain a record of all consent decisions, including those made by the user. This is implemented through the consent history page, where all consent actions, whether initiated by the user or the system, are recorded and displayed. While this supports traceability and accountability, the prototype does not address long-term storage considerations or data portability, which would be necessary in a production system.

Requirement FR10 specifies that the system shall enforce current consent states throughout the application. This is achieved through a centralized state management approach. By using Zustand together with its persistence middleware, all consent states are consistently updated across the system and stored locally using localStorage. However, due to the scope of the prototype, enforcement is limited to a single device and environment, and alternative solutions would be required in real-world scenarios.

In summary, while all functional requirements are implemented and demonstrable within the prototype, several trade-offs and limitations are evident. These primarily relate to usability challenges, scalability concerns, and the absence of a full backend infrastructure. As such, the prototype effectively demonstrates the feasibility of user-centered consent flows but would require further development and evaluation to be suitable for real-world deployment.

6.3 Non-functional Requirements Evaluation

The non-functional requirements specified in Section 4.3 define the quality attributes and design principles that should guide the user-centered consent flow. This Section evaluates how the prototype addresses each requirement against the user testing results presented in Section 6.1.

Requirement NFR1 specifies that the system should be transparent in communicating data processing practices. The prototype achieves this through a structured categorization of purposes (required, functional, and advertising) and an explicit display of data categories and involved parties within the consent interface. User testing resulted in 73% of participants rating transparency at 4 or higher on the Likert scale. However, open-ended feedback indicated gaps in contextual clarity, as detailed in Section 6.1, with participants noting that purpose distinctions and data retention information could be clearer.

Requirement NFR2 specifies that the system should promote usability by minimizing cognitive effort and avoiding overwhelming the user. The prototype employs several usability-focused design decisions, including purpose categorization, dropdown menus for granular controls, and the ability to modify and revoke already given consent in multiple locations. User testing resulted in 87% of participants rating usability at 4 or higher. However, as mentioned in Section 6.1, open-ended feedback revealed concerns about information density and cognitive load, with participants reporting that the interface felt overwhelming when navigating multiple dropdown menus and categories. Participants suggested progressive disclosure, visual summaries, and improved visual design as potential improvements.

Requirement NFR3 specifies the avoidance of dark patterns in consent interfaces. The prototype implements this through unchecked default states, absence of preselected options, and transparent presentation of all consent purposes. User testing resulted in 97% of participants rating avoidance of dark patterns at 4 or higher, the highest satisfaction rating across all categories. No negative feedback was reported for this requirement in the open-ended responses.

Requirement NFR4 specifies that the system should promote user control over consent decisions. The prototype mainly addresses this through the consent dashboard, which enables users to view, modify, and revoke consent decisions and through the ability to modify or revoke consent in multiple locations in the program. The availability of granular controls for non-essential purposes further supports user autonomy. User testing resulted in

87% of participants rating user control at 4 or higher. As noted in Section 6.1, the consent dashboard and consent management features received consistent positive feedback, with participants identifying these as helpful aspects of the system and expressing appreciation for the centralized management of consent decisions.

Requirement NFR5 specifies that the system should implement privacy by default. The prototype achieves this by presenting all processing purposes with unchecked default states, requiring explicit user permission for data sharing. User testing resulted in 87% of participants rating privacy by default at 4 or higher. No critical feedback was reported for this requirement in the open-ended responses.

Requirement NFR6 specifies scalable consent management as the number of data categories, purposes, and third parties increases. The current prototype implements basic scalability through purpose categorization and dropdown menus, which help organize information hierarchically. User testing resulted in 53% of participants rating scalability at 4 or higher, the lowest satisfaction rating across all requirements. This finding aligns with the information density concerns raised in the usability feedback. As noted in Section 6.1, participants experienced increasing difficulty as interface complexity increased, with concerns about data density becoming more pronounced in scenarios with a high number of purposes and third parties. Participants suggested progressive disclosure, visual summaries, and improved information architecture as potential approaches to address these challenges.

Requirement NFR7 specifies that the system should support active user engagement and informed decision-making. The prototype facilitates this through structured presentation of purposes, clear association of data with processing activities, and accessible consent modification interfaces. User testing resulted in 87% of participants rating engagement at 4 or higher. However, as detailed in Section 6.1, open-ended feedback indicated gaps in contextual clarity, with some participants noting confusion about purpose distinctions and the distinction between required and optional consent options.

The prototype demonstrated particular strengths in avoiding dark patterns and supporting user control, usability, privacy by default, and engagement. In contrast, scalability emerged as a challenge, with information density and cognitive load being concerns regarding maintaining usability as information volume grows.

6.4 Reliability and Validity

The reliability and validity of the prototype evaluation and its results are important aspects to consider. Reliability refers to the consistency and stability of the results, while validity concerns whether the evaluation methods accurately measure what they are intended to assess.

The reliability of the analytical evaluation of the prototype, based on the derived requirements, is considered good. This is because the evaluation criteria are grounded in established literature and systematically map the prototype to the derived requirements. The reliability of the user testing is supported by the use of a standardized survey with Likert scale questions, allowing for consistent measurements across all participants. However, the relatively small sample size (fifteen participants) increases sensitivity to individual variation, differences in participants' backgrounds, technical familiarity, and interpretation of questions may introduce variability in the responses. Both the sample size and the use of convenience sampling additionally limit statistical reliability. While the results provide useful indicative trends, they should not be considered definitive or fully generalizable.

The validity of the analytical evaluation is considered strong, as the requirements used as criteria are derived from established literature and principles for user-centered consent design. By directly mapping these requirements to the implemented prototype, the evaluation ensures alignment between theory and implementation. The validity of the user testing is supported by measuring user perceptions of the derived non-functional requirements, such as usability, transparency, and user control. The combination of quantitative Likert scale data and qualitative open-ended feedback further strengthens this. For example, the lower ratings in scalability are consistent with qualitative feedback regarding cognitive load. However, certain limitations affect validity. The user testing primarily measures perceived usability and understanding rather than actual user behavior or long-term interaction patterns. Additionally, the testing was conducted in a controlled setting using a prototype, which may not fully reflect real-world usage scenarios. The validity is therefore limited to this study, as the results may not generalize to larger or more diverse populations or to production-level systems.

Chapter 7

Conclusion and Future Work

This thesis investigated how a user-centered consent flow can be designed and implemented to promote transparency and meaningful user control within digital service platforms. By synthesizing insights from regulatory frameworks, privacy principles, and consent research, a set of requirements was derived and implemented in a web-based prototype. User testing with fifteen participants and a criteria-based assessment was conducted to evaluate how well the prototype addressed these requirements.

The results demonstrate that meaningful user control can be supported by treating consent as a dynamic and continuous process rather than a one-time event. The prototype enabled users to grant, review, modify, and revoke consent through structured interfaces, providing a more continuous approach to consent management. User testing supported this, with the consent dashboard and centralized consent management features receiving consistent positive feedback. Participants valued the ability to easily access and review prior decisions, suggesting that persistent mechanisms for consent oversight address a practical need in current digital environments where users have limited visibility into their data-sharing relationships.

Transparency was pursued through structured presentation of data requests, including explicit display of purposes, data categories, and third parties. However, the relationship between transparency and actual user understanding is not straightforward. While transparency received a positive rating of 73% in user testing, qualitative feedback revealed gaps between information provision and comprehension. Some participants noted that purpose descriptions remained relatively high-level. This suggests that increased information density alone does not guarantee improved understanding. The challenge lies in determining what information is genuinely necessary for informed decision-making and how to present it effectively without overwhelming users. The trade-off between transparency and usability emerged as a central tension: providing comprehensive information about data processing practices tends to increase cognitive burden, yet withholding information undermines the transparency needed for genuine informed consent.

Scalability in consent management refers to the ability of an interface to remain usable and comprehensible as the number of data categories, purposes, and third-party actors increases. This presents a more complex challenge than initial interaction design solutions can address. The prototype attempted to manage complexity through purpose categorization via dropdown menus. However, scalability received the lowest user rating at 53% positive feedback, and participants explicitly described the interface as overwhelming when handling larger amounts of information. Users specifically reported feeling overwhelmed by the number of purposes and dropdowns, as well as the level of detail within them. These observations suggest that scalability issues can be fundamentally rooted in cognitive load rather than mere interface organization. As complexity increases, users face greater cognitive burden regardless of hierarchical structuring. Addressing scalability may therefore require approaches beyond interface design such as simplified data minimization practices that reduce the number of purposes and actors requiring user decisions. The current prototype does not explore scenarios with complex real-world data ecosystems, leaving open questions about whether even well-designed interfaces can adequately support informed decisions when excessive amounts of data are requested.

The prototype demonstrated strong performance in avoiding dark patterns, with 97% of participants rating this aspect positively. This high rating suggests that design choices such as unchecked default options, explicit user actions for consent, and transparent presentation of choices are effective in promoting fair and transparent consent interactions. No negative feedback was reported for this requirement, suggesting that avoiding manipulative design patterns is both technically feasible and valued by users. However, the effectiveness of these measures depends on consistent implementation across all stages of the consent lifecycle. The inclusion of an ”Accept all” button, while practical for usability, introduces a potential point where users might provide consent without full consideration. This illustrates how even well-intentioned usability improvements can create subtle pathways to less informed decisions.

User control, usability, privacy by default, and engagement all received ratings of 87%, indicating general alignment between the design approach and user expectations. The consent dashboard emerged as particularly effective, supporting both practical control (the ability to modify or revoke consent) and awareness of consent decisions, suggesting that ease of access is an important aspect of the prototype. However, the effectiveness of these control mechanisms depends on factors not fully controlled in the prototype design. User willingness to actively engage with consent management interfaces is a prerequisite for these features to have a practical impact. If users perceive consent management as a low priority, even well-designed interfaces may have limited real-world effectiveness.

Overall, this thesis contributes to the understanding of how consent mechanisms can be restructured to better support users in modern data-driven digital environments. The prototype serves as a proof-of-concept demonstrating that user-centered consent flows can be designed by integrating dynamic consent principles, structured interaction design, and usability considerations. The positive user responses to core elements such as consent management, user control, and dark pattern avoidance suggest that the foundational design approach is sound. However, the challenges identified in scalability, transparency-usability balance, and contextual clarity indicate that moving from a simplified prototype to real-world deployment would require significant additional refinement.

7.1 Discussion

While the study provides insights into how transparency and user control can be supported through interface and system design, several limitations must be acknowledged.

Firstly, the evaluation of the prototype is primarily conducted using a criteria-based approach, with limited empirical user testing in the form of convenience sampling. As a result, the findings are largely based on analytical assessment and limited observations of user behavior. Although the evaluation criteria are grounded in established theoretical frameworks, the absence of comprehensive and longitudinal user studies limits the ability to assess long-term real-world usability, user comprehension, and interaction patterns. Consequently, conclusions regarding user experience and effectiveness remain indicative rather than definitive.

Secondly, the prototype is implemented as a proof of concept within a simplified and controlled environment. The system relies on mock data and a lightweight backend solution, and does not include full-scale infrastructure, security mechanisms, or integration with real-world data processing pipelines. While this aligns with the scope of the project, it limits the extent to which the solution is directly applicable in production environments. In particular, aspects such as secure data handling, authentication, and enforcement across distributed systems are not addressed.

Thirdly, avoidance of dark patterns within the prototype scored highly, however, a risk for a manipulative effect of a data exchange offering rewards for data exists. This work, however, focuses on how these exchanges can be made more transparent while supporting user control wherever possible.

Additionally, scalability is considered primarily from a user interaction and information design perspective, rather than from a technical system performance standpoint. Although the prototype demonstrates how consent flows can remain manageable as the complexity of data interactions increases to some extent, it does not evaluate how such mechanisms would perform in large-scale systems with high volumes of users, data interactions, and real-time processing requirements.

Furthermore, the study focuses on a centralized architectural setting, while alternative approaches, such as decentralized blockchain-based consent management, are only discussed at a conceptual level. As a result, the potential benefits and limitations of these architectures are not explored through implementation.

Finally, system requirements and design decisions derive from interpretations of existing literature and regulatory frameworks. While efforts have been made to ensure consistency with established principles, such interpretations may introduce subjectivity, particularly when translating abstract concepts such as transparency and user control into concrete design features.

7.2 Future work

While the prototype demonstrates a feasible approach to designing a user-centered consent flow that supports transparency and meaningful user control, several directions for future work remain.

The evaluation conducted in this thesis is limited to a small-scale user study based on convenience sampling. Future work could prioritize comprehensive empirical validation of the proposed consent flow through larger, more diverse participant groups, as well as longitudinal studies to assess whether the interface genuinely improves user comprehension, reduces consent fatigue, and supports sustained engagement over time. Given the finding from the literature that design improvements do not automatically translate to better user comprehension, such evaluation should specifically measure: (1) user understanding of data processing implications, (2) the effectiveness of comprehension-support mechanisms such as purpose categorization and consent history in fostering informed decisions, and (3) whether users maintain meaningful control and awareness as they interact with the system over extended periods. Due to time constraints, this comprehensive evaluation was not possible; however, it would be particularly relevant for validating the core claims of the prototype.

User testing indicated that scalability in consent management is the prototype's main limitation. As this issue was addressed primarily through improved presentation of complex purposes and choices, future work should explore simplified data minimization practices as a complementary approach. Potential directions include identifying which purposes vendors may rely on under legitimate interest, especially within advertising categories, thereby reducing the number of choices presented to users in the first place. Additionally, adaptive or personalized consent interfaces could be further investigated as a potential mitigation strategy.

Future work could explore deeper integration of the consent flow within real digital service ecosystems, including studying how the proposed model interacts with third-party services, data brokers, and existing infrastructure. Additionally, the prototype should be evaluated against evolving regulatory requirements, including frameworks beyond the General Data Protection Regulation (GDPR) (such as the California Consumer Privacy Act (CCPA), Personal Information Protection and Electronic Documents Act (PIPEDA), and emerging digital regulations in other jurisdictions). This would provide critical insights into the practical applicability and compliance of the proposed approach across different jurisdictions and service contexts.

Central to Privacy by Design (PbD) is the principle of data minimization as an engineering strategy. Future work could investigate how the consent architecture integrates with backend data handling practices. This includes designing systems that respect consent decisions not only at the interface level but throughout data processing pipelines, including data collection scope, storage mechanisms, retention policies, and deletion procedures. The consent system should operate within and actively support broader architectural principles designed to minimize unnecessary data exposure by default, ensuring that user consent decisions are meaningfully enforced in practice.

Future work could investigate adaptive or context-aware consent interfaces that dynamically adjust to different user needs and preferences. For example, the system could adjust the level of detail presented to users based on their privacy literacy, prior behavior, or explicit preferences, offering simplified views for novices while providing granular controls for privacy-conscious users. Alternatively, more advanced consent models, such as AI-assisted or contextual consent management, could support users in making more informed decisions based on contextual cues or historical behavior patterns. This may help balance the trade-off between user transparency and cognitive load. However, such personalization must be carefully designed to avoid creating information asymmetries, inadvertently manipulating users through interface variants, or reinforcing dark pattern-like behaviors. Any personalization approach should be subject to the same transparency and user control principles that guide the current prototype.

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Appendix A

Consent Flow Evaluation Survey

This survey is part of a degree project at KTH Royal Institute of Technology evaluating a prototype for managing user consent in digital systems. The purpose is to gather feedback on usability, transparency, and user control. Please explore the prototype and then answer the following questions based on your experience.

All responses are collected anonymously and will be used solely for academic research purposes. No personally identifiable information is gathered.

Scale

All statements below are rated on a 5-point Likert scale:

- 1 = Strongly disagree
- 2 = Disagree
- 3 = Neutral
- 4 = Agree
- 5 = Strongly agree

Survey Questions

1. **Transparency**
    - The system clearly explains what data is collected and why.
    - The system clearly indicates which parties the data is shared with.
2. **Usability**
    - The interface is easy to use.
    - The amount of information presented is manageable.
3. **Avoidance of Dark Patterns**
    - The interface does not pressure me into making a specific choice.
    - The design is clear and not misleading.
4. **User Control**
    - I can easily view, modify, and withdraw my consent.
    - I feel in control of my personal data.
5. **Privacy by Default**
    - My data is not shared without my explicit permission.
6. **Scalability**
    - The system remains understandable even when multiple data types and purposes are involved.
7. **Engagement**
    - Before consenting, I understand what data will be collected and how it will be used.

Open-ended Questions

8. What did you find most clear or helpful?
9. What did you find confusing or difficult?
10. How could the system be improved?

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